

Gravitational-wave perturbations of a spherical thin shell flat interior, Schwarzschild exterior: junction conditions, scattering coefficients and quasinormal modes

Derivation document accompanying the code in `code/`

Abstract

We derive, from first principles and with symbolic verification, the linearized Israel junction conditions for odd- and even-parity gravitational perturbations of a spherical thin shell that separates a flat interior from a Schwarzschild exterior, and use them to compute reflection/transmission coefficients and quasinormal modes (QNMs). Two shell models are treated: a *truly static* perfect-fluid shell (model C) and a pure domain wall adiabatically frozen at its turning point (model A). The dust shell proposed as “Option B” in the task statement does not exist as a static configuration, and we show why. In the odd sector the junction conditions are model independent and of the scalar form $[\Psi] = 0$, $[d\Psi/dx] = \Delta\Psi$. In the even sector they form a frequency-dependent 2×2 transfer matrix that involves the radial displacement of the shell and the motion of its matter. All results are validated by independent numerical methods and against the literature. In particular we reproduce the Schwarzschild QNMs and reflectivity, and the recent result of Pitre, Schneider & Poisson (2026) that the static fluid shell is dynamically unstable.

Contents

Statement of the model and of the regime of validity	2
1 Background spacetime and shell; the collapse problem	2
1.1 Thin-shell formalism	2
1.2 Spherical shell	3
1.3 The collapse problem and the choice of shell model	3
1.4 Time normalization: interior blueshift	4
2 Perturbation equations	4
2.1 Exterior (Schwarzschild)	5
2.2 Interior (Minkowski)	6
3 Interior solution	6
4 Junction conditions	7
4.1 Linearized matching: method	7
4.2 Odd parity: analytic derivation	7
4.3 Even parity	8
5 Numerical methods	9
5.1 Definition of the scattering coefficients	9
5.2 Method 1: transfer matrix with a continued-fraction exterior basis	10
5.3 Method 2: shooting	10
5.4 QNMs	11
5.5 Schwarzschild black hole and WKB	11

6	Results	11
6.1	Validation checks	11
6.2	Scattering coefficients	13
6.3	Quasinormal modes	14
6.4	Even-parity matter modes and the instability of the static shell	14
7	Discussion	15
8	References and equation map	16
A	Code map	18

Statement of the model and of the regime of validity

- **Model C (primary; truly static).** A perfect-fluid shell, $S^{ab} = \sigma u^a u^b + p(h^{ab} + u^a u^b)$, in *exact* static equilibrium at areal radius $R > 2M$, with $\sigma = (1 - \sqrt{f_R})/(4\pi R)$ and $p = \sigma\kappa$, $\kappa = (1 - \sqrt{f_R})/(4\sqrt{f_R})$ (Sec. 1). In the language of Ipser & Sikivie this is “Option C”, $\tau/\sigma = -\kappa < 0$, with κ fixed by R/M . Perturbations of the shell matter obey a barotropic (polytropic) law $\delta p = \Gamma p \delta\sigma/(\sigma + p)$, and the fiducial adiabatic index is $\Gamma = 2$. The background is exactly static. However, the configuration is *linearly unstable* to even-parity nonradial perturbations. The growth rate is $\text{Im } \omega \simeq 0.5\text{--}0.7 (M/R^3)^{1/2}$ (Pitre, Schneider & Poisson 2026 [PSP26]; reproduced in Sec. 6.4).
- **Model A (adiabatic).** A pure domain wall ($\tau = \sigma$, $S^{ab} = -\sigma h^{ab}$) frozen at its turning point $\dot{R} = 0$, where $\ddot{R} = -(1 + 3\sqrt{f_R})/(2R) < 0$. The background is *not* static; it is momentarily static. The frozen-coefficient approximation is valid for $\omega \gg \omega_{\text{dyn}}$, and the code quantifies its error. The constraint equations are violated at relative order $\omega_{\text{dyn}}/\omega$ and energy conservation at order $(\omega_{\text{dyn}}/\omega)^2$ (Sec. 4.3).
- **Odd parity.** The junction conditions are identical for both models, and indeed for any perfect-fluid shell with $\omega \neq 0$ (Sec. 4.2).
- **Regime.** Real-frequency results (reflection/transmission) describe the static (C) or frozen (A) background. They are physically meaningful for ω well above the instability or collapse rates: $\omega \gg 0.6(M/R^3)^{1/2}$ for C and $\omega \gg \omega_{\text{dyn}}$ for A. Numerical values are given in Table 1. QNMs are eigenvalues of the linearized problem about the stated background.

Units $G = c = 1$. Frequencies are quoted as ωM , the time dependence is $e^{-i\omega t}$, and t is the Schwarzschild (Killing) time of the exterior.

1 Background spacetime and shell; the collapse problem

1.1 Thin-shell formalism

Let Σ be the timelike world-tube of the shell with intrinsic coordinates $y^a = (\tau, \theta, \phi)$, tangent vectors $e_a^\alpha = \partial x^\alpha / \partial y^a$, and unit normal n_α pointing from the interior (−) to the exterior (+). The induced metric and extrinsic curvature are

$$h_{ab} = g_{\alpha\beta} e_a^\alpha e_b^\beta, \quad K_{ab} = e_a^\alpha e_b^\beta \nabla_\alpha n_\beta = -n_\mu (\partial_a e_b^\mu + \Gamma_{\nu\rho}^\mu e_a^\nu e_b^\rho), \quad (1.1)$$

[Ipser & Sikivie 1984 (IS84) Eqs. (2.1)–(2.3); Poisson 2004 (P04) Secs. 3.1, 3.7]. The junction conditions are the continuity of the induced metric and the Lanczos–Israel condition,

$$[h_{ab}] = 0, \quad [K_{ab}] - h_{ab}[K] = -8\pi S_{ab}, \quad [X] \equiv X|_+ - X|_-, \quad (1.2)$$

(P04 Eqs. 3.49 and 3.56; IS84 Eq. 2.6). They imply the conservation law $D_b S^{ab} = 0$ when the two sides are vacuum (IS84 Eq. 2.8a).

1.2 Spherical shell

Birkhoff's theorem and regularity at $r = 0$ fix the interior to be Minkowski (interior_ondas_cascara, Sec. 1), $ds_-^2 = -dT^2 + dr^2 + r^2 d\Omega^2$. The exterior is Schwarzschild, $ds_+^2 = -f dt^2 + f^{-1} dr^2 + r^2 d\Omega^2$ with $f = 1 - 2M/r$ (IS84 Eq. 3.1). For a shell $r = R(\tau)$ with $\alpha = (1 + \dot{R}^2)^{1/2}$ and $\beta = (f(R) + \dot{R}^2)^{1/2}$ (IS84 Eq. 3.4), the nonvanishing mixed components of the extrinsic curvature are (IS84 Eqs. 3.5–3.6; P04 Eq. 3.65)

$$K^\theta_\theta|_+ = \frac{\beta}{R}, \quad K^\theta_\theta|_- = \frac{\alpha}{R}, \quad K^\tau_\tau|_+ = \frac{\ddot{R} + M/R^2}{\beta}, \quad K^\tau_\tau|_- = \frac{\ddot{R}}{\alpha}. \quad (1.3)$$

With $S^a_b = \text{diag}(-\sigma, p, p)$, Eq. (1.2) gives

$$\sigma = -\frac{1}{4\pi} [K^\theta_\theta] = \frac{\alpha - \beta}{4\pi R}, \quad p = \frac{1}{8\pi} ([K^\tau_\tau] + [K^\theta_\theta]). \quad (1.4)$$

These are equivalent to IS84's equations of motion, Eqs. (3.7a,b), written with the tension $\tau = -p$:

$$(\alpha + \beta)\ddot{R} = -\frac{\alpha M}{R^2} - \frac{2\tau}{\sigma} \frac{\alpha\beta(\alpha + \beta)}{R}, \quad (\alpha - \beta)\ddot{R} = -\frac{\alpha M}{R^2} + 4\pi(\sigma - 2\tau)\alpha\beta. \quad (1.5)$$

Squaring the first of Eqs. (1.4) gives the mass formula $M = \frac{1}{2}(\alpha + \beta)4\pi\sigma R^2$ (IS84 Eq. 3.8). At a moment with $\dot{R} = 0$ this becomes

$$4\pi\sigma R = 1 - \sqrt{f_R}, \quad \text{equivalently} \quad M = 4\pi\sigma R^2(1 - 2\pi\sigma R), \quad (1.6)$$

which is IS84 Eq. (3.9). The code checks it to machine precision (Table 1). Here and below $f_R \equiv f(R)$.

1.3 The collapse problem and the choice of shell model

Setting $\dot{R} = \ddot{R} = 0$ in Eq. (1.5) requires

$$\frac{M}{R^2} + \frac{2\tau}{\sigma} \frac{f_R + \sqrt{f_R}}{R} = 0 \quad \implies \quad \frac{\tau}{\sigma} = -\kappa, \quad \kappa = \frac{M}{2R\sqrt{f_R}(1 + \sqrt{f_R})} = \frac{1 - \sqrt{f_R}}{4\sqrt{f_R}} > 0. \quad (1.7)$$

A static shell therefore needs *negative tension*, i.e. positive surface pressure $p = \kappa\sigma$. This reproduces P04 Eq. (3.81), $p = (1 - M/R - \sqrt{f_R})/(8\pi R\sqrt{f_R})$. We now evaluate the options listed in the task.

Option B (static dust shell) does not exist. For dust, $\tau = 0$, and the first of Eqs. (1.5) gives $(\alpha + \beta)\ddot{R} = -\alpha M/R^2 < 0$ for any $\dot{R}$. As IS84 state after their Eq. (3.7a), “ $\ddot{R}$ is always negative if $\tau \geq 0$ ”. The same conclusion follows from P04 Eq. (3.70), $M = m(1 + \dot{R}^2)^{1/2} - m^2/2R$ with $m = 4\pi\sigma R^2$ conserved: a dust shell can only be *momentarily* static, at the turning point. There it obeys the same relation (1.6) as the domain wall, and $\ddot{R} = -M/[R^2(1 + \sqrt{f_R})]$. The formula $M = 4\pi\sigma R^2\sqrt{f_R}$ quoted in the task statement does not follow from the junction conditions. It coincides with neither $M = 4\pi\sigma R^2(1 - 2\pi\sigma R)$ nor the static relation $4\pi\sigma R = 1 - \sqrt{f_R}$. We therefore do not implement Option B.

Option C (static fluid shell): chosen as primary model (model C). Equation (1.7) shows that κ is *not* a free parameter: for static equilibrium it is fixed by R/M (Table 1). The family is parametrized by compactness. The dominant energy condition $p \leq \sigma$ requires $\kappa \leq 1$, i.e. $\sqrt{f_R} \geq 1/5$ or $R \geq 25M/12 \simeq 2.083M$. For perturbations one must also specify how the shell's pressure responds, $\delta p = v_s^2 \delta\sigma$. We use the polytropic law of PSP26 (their Eqs. 1.1, 2.5–2.6), $\delta p = \Gamma p/(\sigma + p) \delta\sigma$, i.e. $v_s^2 = \Gamma\kappa/(1 + \kappa)$, with fiducial $\Gamma = 2$. Our σ is PSP26's energy density μ . Causality $v_s^2 \leq 1$ holds for $\Gamma = 2$ iff $\kappa \leq 1$. Our model-C results are therefore restricted to $R \geq 2.1M$, except where stated.

Table 1: Background quantities (from `results/validation.json`). The column $(M/R^3)^{1/2}$ sets the scale of the model-C instability rate ($\simeq 0.6$ times this) and of the matter modes. ω_{dyn} is the domain-wall collapse rate in exterior time.

R/M	σM	pM	$\kappa = p/\sigma$	$v_s^2(\Gamma=2)$	I&S (3.7a) resid.	$\ddot{R}_{\text{wall}}M$	$\omega_{\text{dyn}}M$	$(M/R^3)^{1/2}M$
2.2	0.02527	1.463e-02	0.5792	0.7335	0.00e+00	-0.4328	0.1337	0.3065
3.0	0.01121	2.052e-03	0.1830	0.3094	1.39e-17	-0.4553	0.2249	0.1925
6.0	0.00243	1.367e-04	0.0562	0.1064	0.00e+00	-0.2875	0.1787	0.0680
10.0	0.00084	2.479e-05	0.0295	0.0573	1.73e-18	-0.1842	0.1214	0.0316

Option A (domain wall, adiabatic): implemented as model A. For $\tau = \sigma$ and $\dot{R} = 0$, Eqs. (1.3)–(1.4) give $[K^\tau_\tau] = [K^\theta_\theta] = -4\pi\sigma$, and hence

$$\ddot{R} = -\frac{M}{R^2(1+\sqrt{f_R})} - \frac{2\sqrt{f_R}}{R} = -\frac{1+3\sqrt{f_R}}{2R} \quad (\text{proper time}). \quad (1.8)$$

The first form is IS84 Eq. (3.7a) with $\alpha = 1$, $\beta = \sqrt{f_R}$. The symbolic derivation obtains the same $\ddot{R}$ independently, from the angular Israel equation at the frozen instant. The background is momentarily static. We define the dynamical frequency in exterior time as $\omega_{\text{dyn}} = \sqrt{f_R}(|\dot{R}|/R)^{1/2}$; the frozen approximation requires $\omega \gg \omega_{\text{dyn}}$.

Option D (Schwarzschild/de Sitter interior). Not implemented. It changes the interior problem, and Pani et al. (2009) have treated its de Sitter version. Our model C is the $\Sigma \neq 0$, Minkowski-interior member of the same family of thin-shell “gravastars”.

Why both static options end up dynamical. Model A collapses on the timescale ω_{dyn}^{-1} . Model C has an exact static background, but it possesses an unstable even-parity “matter mode”. PSP26 found this mode, and we find it independently (Sec. 6.4), with growth rate $\simeq 0.6(M/R^3)^{1/2}$. In both cases real-frequency scattering is an idealization valid for ω large compared with these rates. Wave-mode QNMs have $\omega \sim 1/R$, which is larger than $(M/R^3)^{1/2}$ for all $R > 2M$ by a factor $(R/M)^{1/2}$.

1.4 Time normalization: interior blueshift

Continuity of h_{ab} at a static shell requires $dT = \sqrt{f_R} dt$ (P04 Eq. 3.75). A monochromatic perturbation $e^{-i\omega t}$ outside is therefore $e^{-i\nu T}$ inside, with

$$\nu = \frac{\omega}{\sqrt{f_R}}. \quad (1.9)$$

The regular interior solution is $\hat{j}_\ell(\nu r)$, not $\hat{j}_\ell(\omega r)$ as written in the prior notes and in the task statement. This is essential near $R \rightarrow 2M$, where $\nu \gg \omega$. PSP26 (their $w = F^{-1/2}\omega$) adopt the same normalization. It is convenient to define a *global tortoise coordinate*

$$x = r_* = r + 2M \ln(r/2M - 1) \quad (r > R), \quad x = r_*(R) + (r - R)/\sqrt{f_R} \quad (r < R), \quad (1.10)$$

in which both sides obey $-\partial_t^2 \Psi + \partial_x^2 \Psi - V\Psi = 0$, with $V_{\text{int}} = f_R \ell(\ell+1)/r^2$ (Fig. 1).

2 Perturbation equations

We follow Martel & Poisson (2005) [MP05] throughout. Spherical-harmonic labels are omitted. The odd-parity harmonics are $X_A = -\varepsilon_A^B D_B Y$ and $X_{AB} = -\frac{1}{2}(\varepsilon_A^C D_B + \varepsilon_B^C D_A) D_C Y$ (MP05 Eqs. 3.2, 3.7). The even-parity ones are $Y_A = D_A Y$ and $Y_{AB} = [D_A D_B + \frac{1}{2}\ell(\ell+1)\Omega_{AB}]Y$ (MP05 Eqs. 3.1, 3.6). We use $\lambda = \ell(\ell+1)$ and $\mu = (\ell-1)(\ell+2)$, and Regge–Wheeler (RW) gauge on each side: $h_2 = 0$ (odd) and $j_a = G = 0$ (even) (MP05 Secs. IV.A, V.A).

Effective potentials: exterior V_{RW}, V_Z ; flat interior $f_R \ell(\ell+1)/r^2$; shell = repulsive δ -coupling (odd: $\Delta = \sqrt{f_R}(1 - \sqrt{f_R})/R$)

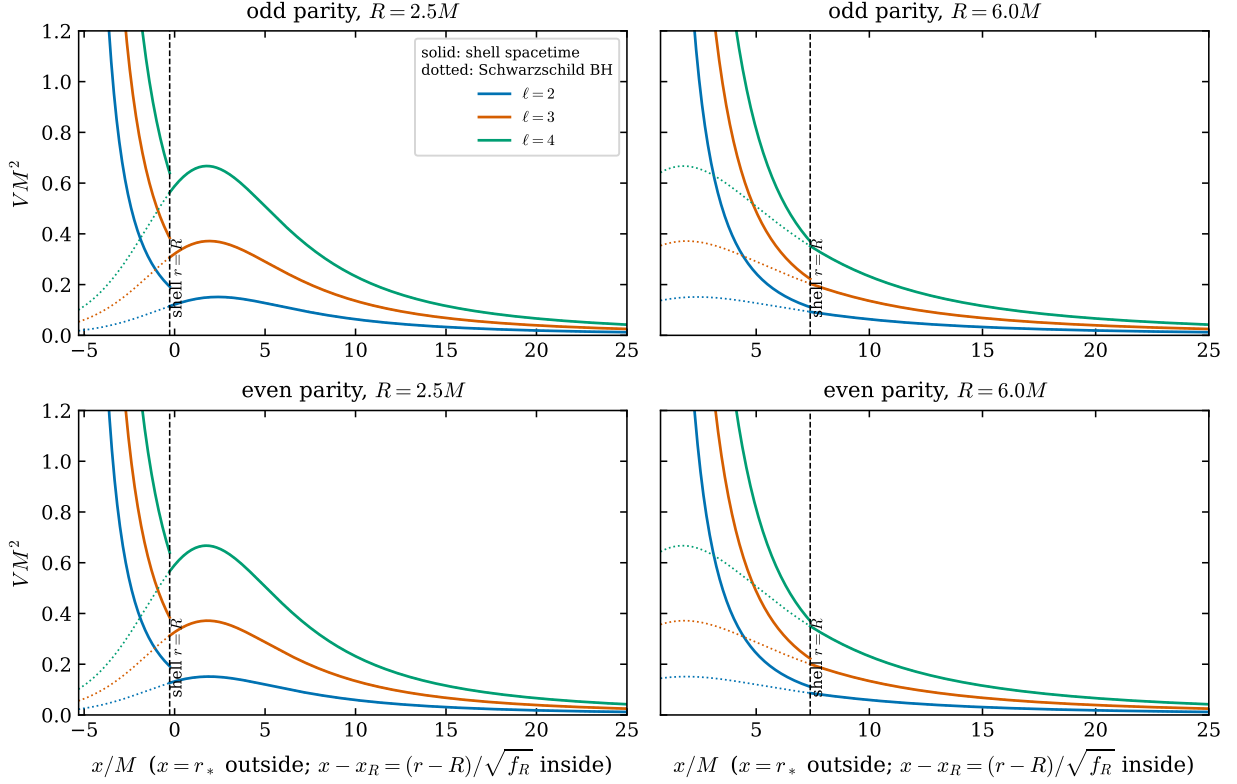


Figure 1: Effective potentials in the global tortoise coordinate (1.10) for $\ell = 2, 3, 4$. Outside: V_{RW} (RW57 Eq. 25; MP05 Eq. 5.15) and V_Z (Zerilli 1970 Eq. 5; MP05 Eq. 4.26). Inside: $f_R \ell(\ell+1)/r^2$. Dotted lines show the Schwarzschild continuation to the horizon. The odd-parity shell adds a repulsive δ -function of strength Δ_{odd} , Eq. (4.6). The even-parity coupling is a frequency-dependent 2×2 matrix.

2.1 Exterior (Schwarzschild)

Odd parity. $p_{aB} = h_a X_B$ with $h_a = (h_t, h_r)$ (MP05 Eq. 5.2). The Cunningham–Price–Moncrief function

$$\Psi_{\text{odd}} = \frac{2r}{\mu} \varepsilon^{ab} \left(\partial_a h_b - \frac{2}{r} r_a h_b \right) = \frac{2r}{\mu} \left(\partial_r h_t - \partial_t h_r - \frac{2}{r} h_t \right) \quad (2.1)$$

(MP05 Eq. 5.13, with $\varepsilon_{tr} = 1$) obeys the Regge–Wheeler equation (MP05 Eqs. 5.14–5.15; RW57 Eq. 25)

$$\frac{d^2 \Psi}{dr_*^2} + [\omega^2 - V_{\text{RW}}(r)] \Psi = 0, \quad V_{\text{RW}} = f \left[\frac{\lambda}{r^2} - \frac{6M}{r^3} \right]. \quad (2.2)$$

The vacuum field equations, MP05 Eqs. (5.8), (5.9), (5.18), can be inverted as

$$h_t = \frac{f}{2} \partial_r (r \Psi_{\text{odd}}), \quad h_r = \frac{r}{2f} \partial_t \Psi_{\text{odd}} = -\frac{i\omega r}{2f} \Psi_{\text{odd}}. \quad (2.3)$$

The second relation is MP05 Eq. (5.18), $\Psi_{\text{RW}} = r^a h_a / r = \frac{1}{2} \partial_t \Psi_{\text{odd}}$, in vacuum. The first follows from $P = \nabla_a h^a = 0$ (MP05 Eq. 5.9). *Verification: symbolic/vacuum_relations.py* linearizes the Einstein tensor by brute force for a generic axisymmetric harmonic, keeping λ symbolic. It checks that Eqs. (2.3) satisfy all odd-parity field equations identically when Ψ obeys (2.2), and that (2.1) reproduces Ψ (log in `results/log_vacuum_relations.txt`).

Even parity. $p_{ab} = h_{ab} Y$ with $h_{tt} = f H_0$, $h_{tr} = H_1$, $h_{rr} = H_2/f$, and $p_{AB} = r^2 K \Omega_{AB} Y$ (MP05 Eqs. 4.1–4.3). The trace-free angular equation gives $H_0 = H_2 \equiv H$. The Zerilli–Moncrief function

(MP05 Eqs. 4.23–4.24)

$$\Psi_{\text{even}} = \frac{2r}{\lambda} \left[K + \frac{2}{\Lambda} (r^a r^b h_{ab} - r r^a \nabla_a K) \right] = \frac{2r}{\lambda} \left[K + \frac{2f}{\Lambda} (H - rK') \right], \quad \Lambda = \mu + \frac{6M}{r}, \quad (2.4)$$

obeys the Zerilli equation, i.e. Eq. (2.2) with $V_{\text{RW}} \rightarrow V_Z$,

$$V_Z = f \frac{2n^2(n+1)r^3 + 6n^2Mr^2 + 18nM^2r + 18M^3}{r^3(nr + 3M)^2}, \quad n = \frac{\mu}{2}, \quad (2.5)$$

(Zerilli 1970 Eq. 5). This equals f times MP05 Eq. (4.26); the identity is checked symbolically. The code derives the first-order system (the tr , $t\theta$ and $r\theta$ components of the linearized Einstein equations; it agrees with Zerilli 1970's system) and the algebraic identity (from G_{rr}). It then inverts (2.4) and obtains the reconstruction

$$K = f \Psi' + \frac{\mu(\mu+2)r^2 + 6\mu Mr + 24M^2}{2r^2(\mu r + 6M)} \Psi, \quad (2.6)$$

$$H_1 = -i\omega \left[r \Psi' + \frac{\mu r^2 - 3\mu Mr - 6M^2}{(r-2M)(\mu r + 6M)} \Psi \right], \quad (2.7)$$

$$H = f \frac{\mu r^2 - 3\mu Mr - 6M^2}{(r-2M)(\mu r + 6M)} \Psi' + \left[\frac{Q(r)}{2r^2(r-2M)(\mu r + 6M)^2} - \frac{\omega^2 r}{f} \right] \Psi, \quad (2.8)$$

$$Q(r) = \mu^2(\mu+2)r^4 - 2M\mu^2(\mu-1)r^3 - 12M^2\mu(\mu-3)r^2 - 72M^3(\mu-1)r - 144M^4,$$

where $' = d/dr$. Substituting (2.6)–(2.8) into all six independent linearized Einstein equations gives zero identically once Ψ obeys the Zerilli equation. The check is done exactly, in rational arithmetic, at random rational values of (r, M, ω, λ) . Substituting them into (2.4) returns Ψ .

Chandrasekhar transformation. The map $Z = [\mu(\mu+2) + 72M^2f/(r(\mu r + 6M))]\Psi_{\text{RW}} + 12M d\Psi_{\text{RW}}/dr_*$ (Chandrasekhar 1983, Sec. 4.26; used by Pani et al. 2009 Sec. III.B) takes RW solutions into Zerilli solutions for all ω . It is verified symbolically in `symbolic/exterior_series.py`. Outgoing solutions map to outgoing ones, with amplitude factor $\mu(\mu+2) + 12iM\omega$.

2.2 Interior (Minkowski)

With $M = 0$ both potentials reduce to λ/r^2 , so both master functions obey (`interior_ondas_cascara` Eqs. 6–9)

$$\Psi'' + \left[\nu^2 - \frac{\lambda}{r^2} \right] \Psi = 0 \quad (0 < r < R), \quad (2.9)$$

with ν the interior-time frequency (1.9). The reconstruction formulas above hold with $M = 0$ and $\omega \rightarrow \nu$, since the interior is written in its own Minkowski time T .

3 Interior solution

The Riccati–Bessel functions $\hat{j}_\ell(z) = z j_\ell(z)$ and $\hat{y}_\ell(z) = z y_\ell(z)$ (Abramowitz & Stegun 1972 Sec. 10.3; `interior_ondas_cascara` Eqs. 10, 26) solve (2.9) with $z = \nu r$. Only

$$\Psi_{\text{int}} = A \hat{j}_\ell(\nu r), \quad \hat{j}_\ell(z) \rightarrow \frac{z^{\ell+1}}{(2\ell+1)!!} \quad (z \rightarrow 0), \quad (3.1)$$

is regular at the centre (`interior_ondas_cascara` Eqs. 11–13). The two parities share this profile; they differ only through the junction conditions. For the *open-interior* scattering problem (Sec. 5.1) we also use the Riccati–Hankel functions ($e^{-i\omega t}$ convention)

$$\hat{h}_\ell^{\text{out}} = -\hat{y}_\ell + i\hat{j}_\ell \sim e^{+i(z-\ell\pi/2)}, \quad \hat{h}_\ell^{\text{in}} = -\hat{y}_\ell - i\hat{j}_\ell \sim e^{-i(z-\ell\pi/2)}, \quad (3.2)$$

with Wronskian $\hat{j}_\ell \hat{y}_\ell' - \hat{j}_\ell' \hat{y}_\ell = 1$ (DLMF 10.50.1), so that $\text{Im}(\hat{h}^{\text{in}*} \hat{h}^{\text{in}}) = -1$ exactly. The code evaluates $\hat{j}_\ell$ both through `scipy.special.spherical_jn` (complex argument) and by direct DOP853 integration of (2.9) from a power-series start. The two agree to 10^{-10} (`interior.regular_numerical`).

4 Junction conditions

4.1 Linearized matching: method

The perturbed metric on each side is written in that side's own RW gauge. RW gauge fixes the bulk gauge freedom completely for $\ell \geq 2$, but it is not adapted to the shell. The shell's world-tube is therefore described on each side by a perturbed embedding (cf. Pani et al. 2009 Sec. III.C and App. A)

$$x_{\pm}^{\mu}(y) = X_{0\pm}^{\mu}(y) + \epsilon Z_{\pm}^{\mu}(y), \quad Z_+ = (0, \xi_+ Y, 0, 0), \quad Z_- = (\zeta Y, \xi_- Y, \eta \partial_{\theta} Y, 0) [+ \chi X^{\phi} \text{ (odd)}], \quad (4.1)$$

where $y^a = (\tau, \theta, \phi)$ are chosen to coincide with the exterior ones. $\xi_{\pm}$ are the radial displacements of the shell in the two RW gauges. ζ, η (χ) describe the relative time/angular identification (twist) of the two sides. The unit normal is solved to first order from $n_{\mu} e_a^{\mu} = 0$ and $g^{\mu\nu} n_{\mu} n_{\nu} = 1$. The induced metric and (1.1) are then expanded to $O(\epsilon)$ by brute force, including the displacement of the background fields, $Q(X_0 + \epsilon Z) = Q(X_0) + \epsilon Z^{\mu} \partial_{\mu} Q$. The result is projected on harmonics. The bulk metric functions are replaced by the master functions through Eqs. (2.3) and (2.6)–(2.8). The matter of the shell is perturbed as $\sigma \rightarrow \sigma + \delta\sigma Y$, $p \rightarrow p + v_s^2 \delta\sigma Y$, $u_A = U D_A Y$ (even) or $U X_A$ (odd). For the domain wall, $S_{ab} = -\sigma h_{ab}$ with σ constant (IS84 Eq. 2.12). The equations (1.2) become a linear system,

$$A(\omega) \mathbf{x} = B(\omega) \begin{pmatrix} \Psi_-(R) \\ \Psi'_-(R) \end{pmatrix}, \quad \mathbf{x} = (\text{shell variables}, \Psi_+(R), \Psi'_+(R)), \quad (4.2)$$

whose last two components define the transfer matrix T : $(\Psi_+, \Psi'_+)^T = T(\Psi_-, \Psi'_-)^T$. The whole construction is implemented in `symbolic/derive_junction.py`. That script also verifies the background Israel equations: they vanish identically for Poisson's σ , p and yield Eq. (1.8) for the wall. It exports the exact matrices A, B as the Python modules `shellgw/junction-{odd,even}-{A,C}.py`.

4.2 Odd parity: analytic derivation

Odd perturbations do not move the shell radially. e_a and the normal are unperturbed except for the twist χ , and $n_{\alpha} = f^{-1/2} \partial_{\alpha} r$ remains exact to first order because g^{rr} is not perturbed. From (1.1), $K_{ab} = -N \Gamma^r_{ab}$ with $N = f^{-1/2}$. The first-order Christoffel symbols $\delta \Gamma^r_{tA} = \frac{f}{2} (\partial_t h_r - \partial_r h_t) X_A$ and $\delta \Gamma^r_{AB} = f h_r X_{AB}$ then give

$$\delta K_{tA} = \frac{\sqrt{f}}{2} (\partial_r h_t - \partial_t h_r) X_A, \quad \delta K_{AB} = -\sqrt{f} h_r X_{AB}. \quad (4.3)$$

The first is Eq. (12) of `burbuja_matching_israel`. With Eq. (2.1), $\partial_r h_t - \partial_t h_r = \mu \Psi / (2r) + 2h_t / r$. We convert to proper time on Σ ($e_{\tau} = f_R^{-1/2} \partial_t$ outside, ∂_T inside) and define the common quantity $\mathcal{H} \equiv \delta h_{\tau A}$, the coefficient of X_A . Then

$$\text{outside: } \mathcal{H} = \frac{h_t^+}{\sqrt{f_R}}, \quad \delta K_{\tau A}^+ = \frac{\mu \Psi_+}{4R} + \frac{\sqrt{f_R} \mathcal{H}}{R}; \quad \text{inside: } \mathcal{H} = h_T^-, \quad \delta K_{\tau A}^- = \frac{\mu \Psi_-}{4R} + \frac{\mathcal{H}}{R}. \quad (4.4)$$

Continuity of $\delta h_{\tau A}$ is built in by using the single symbol $\mathcal{H}$. Continuity of δh_{AB} is automatic, since $h_2 = 0$ on both sides and hence $\chi = 0$. Three equations remain:

1. The *AB Israel equation*. Since $\delta h_{AB} = 0$ and $u_A u_B = O(\epsilon^2)$, $\delta S_{AB}^{\text{odd}} = 0$, and so $[\sqrt{f} h_r] = 0$. With (2.3) on each side ($h_r^- = -i\nu R \Psi_- / 2$), this gives exactly

$$\Psi_+(R) = \Psi_-(R). \quad (4.5)$$

2. The *τA Israel equation*. With $\delta S_{\tau A} = -(\sigma + p)U + p \mathcal{H}$ and the trace of (1.2), $[K] = 4\pi(2p - \sigma)$, we get $[\delta K_{\tau A}] = -4\pi\sigma \mathcal{H} + 8\pi(\sigma + p)U$. By (4.4) the left side equals $\mu[\Psi]/(4R) + (\sqrt{f_R} - 1)\mathcal{H}/R$,

and $(\sqrt{f_R} - 1)/R = -4\pi\sigma$. Hence $\mu[\Psi]/(4R) = 8\pi(\sigma + p)U$, and with (4.5), $U = 0$: the shell matter does not move. This is consistent with the odd part of $D_b S^{ab} = 0$, which gives $\partial_\tau(u_A) = 0$, hence $U = 0$ for $\omega \neq 0$. The background trace $[K]^{(0)}$ identified as “the remaining step” in `burbuja_matching_israel` Eq. (13) combines with $\delta S_{\tau A}$ into $[K]^{(0)} - 8\pi p = -4\pi\sigma$. The p dependence, and with it the $\vec{R}$ dependence of the domain wall, *cancels identically*.

3. *Continuity of $\mathcal{H}$* . From (2.3), $\mathcal{H} = \frac{1}{2}(r\Psi_-)'|_R = \frac{\sqrt{f_R}}{2}(r\Psi_+)'|_R$.

With (4.5), item 3 gives

$$\boxed{\Psi_+ = \Psi_-, \quad \sqrt{f_R}\Psi'_+ - \Psi'_- = \frac{1 - \sqrt{f_R}}{R}\Psi = 4\pi\sigma\Psi} \iff \left[\frac{d\Psi}{d\ell}\right] = 4\pi\sigma\Psi, \quad (4.6)$$

with ℓ the proper radial distance. In the global tortoise coordinate (1.10), $[d\Psi/dx] = \Delta_{\text{odd}}\Psi$ with

$$\Delta_{\text{odd}} = \frac{\sqrt{f_R}(1 - \sqrt{f_R})}{R} = 4\pi\sigma\sqrt{f_R} > 0, \quad (4.7)$$

i.e. the shell acts as a *repulsive δ -function potential* $\Delta_{\text{odd}}\delta(x - x_R)$. This is precisely the schematic form anticipated in the task, with Δ independent of ω and of the shell’s equation of state. The transfer matrix is $\mathsf{T}_{\text{odd}} = \begin{pmatrix} 1 & 0 \\ (1 - \sqrt{f_R})/(R\sqrt{f_R}) & 1/\sqrt{f_R} \end{pmatrix}$, and $\det \mathsf{T}_{\text{odd}} = f_R^{-1/2}$ (flux conservation, Sec. 5.1).

Checks. (i) The brute-force symbolic derivation, with the twist χ and the velocity U kept as unknowns, reproduces (4.6) for model C (4 equations, 4 unknowns, $\chi = U = 0$) and model A. For model A the system is overdetermined (4 equations, 3 unknowns) but consistent, with rank 3; the difference between the two models’ transfer matrices is 10^{-15} . (ii) (4.6) agrees with PSP26 Eq. (5.23)/(97), $F^{1/2}\eta_{\text{out}} - \eta_{\text{in}} + F^{1/2} - 1 = 0$ with $\eta = R\psi'/\psi$. The phase factors of their retarded-time slicing cancel because $w = F^{-1/2}\omega$. (iii) It agrees with Pani et al. (2009) Eqs. (3.29) and (A35), $[h_0] = 0$ and $[\sqrt{h}h_1] = 0$. In their $\Sigma = 0$ model h is continuous and Ψ, Ψ' are continuous; for $\Sigma \neq 0$ their Regge–Wheeler function $\Psi_{\text{RW}} \propto \sqrt{h}h_1$ jumps, whereas the CPM function (2.1) is continuous. (iv) As $M \rightarrow 0$ ($\sigma \rightarrow 0$), $\mathsf{T} \rightarrow \mathbb{I}$: the inviscid/transparent limit of Boyanov et al. (2024), continuity of Ψ and Ψ' .

4.3 Even parity

Induced metric. The first-order induced metric computed from the two sides is (harmonic coefficients of Y , $D_A Y$, $\Omega_{AB} Y$ and Y_{AB})

$$\begin{aligned} \text{outside: } \delta h_{\tau\tau} &= H_+ - \frac{2M}{R(R-2M)}\xi_+, & \delta h_{\tau A} &= 0, & \delta h_{AB}^{\text{tr}} &= R^2 K_+ + 2R\xi_+, & \delta h_{AB}^{\text{tf}} &= 0, \\ \text{inside: } \delta h_{\tau\tau} &= H_- + 2i\nu\zeta, & \delta h_{\tau A} &= -\zeta - i\nu R^2\eta, & \delta h_{AB}^{\text{tr}} &= R^2 K_- - \lambda R^2\eta + 2R\xi_-, & \delta h_{AB}^{\text{tf}} &= 2R^2\eta. \end{aligned} \quad (4.8)$$

The first junction condition therefore forces $\eta = \zeta = 0$ (no relative twist of the two identifications) and leaves

$$H_+ - \frac{2M}{R(R-2M)}\xi_+ = H_-, \quad RK_+ + 2\xi_+ = RK_- + 2\xi_-. \quad (4.9)$$

These agree with Pani et al. (2009) Eqs. (A16) and (A20), $[[2hz + aK]] = 0$ and $[[x]] = [[w]] = [[y]] = 0$. The radial displacement of the shell differs between the two RW gauges ($\xi_+ \neq \xi_-$). This is why the even sector cannot be reduced to a scalar jump condition.

Israel equations. The four even components ($\tau\tau$, τA , trace and trace-free AB) of $[\delta K_{ab} - \delta h_{ab}K - h_{ab}\delta K] = -8\pi\delta S_{ab}$ involve $(H_{\pm}, H'_{\pm}, K_{\pm}, K'_{\pm}, H_{1\pm}, \xi_{\pm})$ and the matter variables. For model C,

$$\delta S_{\tau\tau} = \delta\sigma - \sigma\delta h_{\tau\tau}, \quad \delta S_{\tau A} = -(\sigma + p)U + p\delta h_{\tau A}, \quad \delta S_{AB}^{\text{tr}} = v_s^2\delta\sigma R^2 + p\delta h_{AB}^{\text{tr}}, \quad \delta S_{AB}^{\text{tf}} = p\delta h_{AB}^{\text{tf}}. \quad (4.10)$$

For model A, $\delta S_{ab} = -\sigma\delta h_{ab}$. With Eqs. (2.6)–(2.8) on each side ($M \rightarrow 0$, $\omega \rightarrow \nu$ inside), the system (4.2) has the following size.

- Model C: 8 equations for the 8 unknowns $(\xi_+, \xi_-, \zeta, \eta, \delta\sigma, U, \Psi_+, \Psi'_+)$. It has full rank, and its solution defines $\mathsf{T}_{\text{even}}^{\text{C}}(\omega; \ell, R/M, \Gamma)$.
- Model A: 8 equations for 6 unknowns (no matter variables). See below.

The individual first-order extrinsic-curvature components are lengthy. They are generated and stored exactly by `symbolic/derive_junction.py` and exported to `shellgw/junction_even-{C,A}.py`.

Structure of the result. In contrast to odd parity,

- Ψ is *discontinuous*. T_{even} is a full 2×2 matrix that depends on frequency (through ω^2 terms in H and through the shell's inertia) and on the shell equation of state through v_s^2 . The schematic form $[\Psi] = 0$, $[\partial_x \Psi] = \Delta \Psi$ does *not* hold. As anticipated in the task, the even problem is a genuine 2×2 matching, with the shell's radial motion $\xi_{\pm}$ and its density and velocity perturbations eliminated.
- $\det \mathsf{T}_{\text{even}}^{\text{C}} = f_R^{-1/2}$ exactly. In the code this holds to 10^{-12} in double precision and 10^{-15} in 40-digit arithmetic, for all ω , ℓ , R and Γ tested. For real ω the matrix conserves the flux J , so $\mathcal{R} + \mathcal{T} = 1$ (Sec. 6.1).
- Transparent limit: $\mathsf{T}_{\text{even}} \rightarrow \mathbb{I}$ as $M \rightarrow 0$ at fixed R (i.e. $\sigma, p \rightarrow 0$), for both models, to 4×10^{-9} at $M = 10^{-8}$.
- The matrix has poles at the frequencies of the *matter modes* of the shell, which are absent in odd parity. For model C their QNM counterparts, including a purely imaginary unstable mode, reproduce PSP26 quantitatively (Sec. 6.4).

Model A: an adiabatic, not exact, matching. For an exact solution of the moving-shell problem, the $\tau\tau$ and τA Israel equations follow from the others by the Codazzi identity $D_b([K^b_a] - \delta_a^b K) = 0$ together with $D_b S^{ab} = 0$. Accordingly, the odd-parity model-A system, although overdetermined, is exactly consistent. In the even sector the frozen, monochromatic ansatz is *not* an exact solution at the turning point, because $\ddot{T}_0 \propto \ddot{R}^2 \neq 0$ there even though $\dot{R} = \dot{T}_0 = 0$. The two constraint rows are then violated. Indeed $\text{rank } \mathbf{A} = 6$ while $\text{rank}[\mathbf{A}|\mathbf{B}] = 8$. We solve the six evolution equations (continuity and angular Israel equations) and monitor the constraints. Their relative violation decreases as $\omega_{\text{dyn}}/\omega$, while $|\mathcal{R} + \mathcal{T} - 1|$ and $|\det \mathsf{T} \sqrt{f_R} - 1|$ decrease as $(\omega_{\text{dyn}}/\omega)^2$ (Table 2, Fig. 11). This is the precise sense in which Option A is a high-frequency (adiabatic) approximation. Its even-parity results should be read as $O(\omega_{\text{dyn}}/\omega)$ -accurate.

5 Numerical methods

5.1 Definition of the scattering coefficients

Because the interior is regular and nothing is absorbed, the *physical* (closed) system reflects everything: $\Psi_+ \rightarrow B_{\text{in}}e^{-i\omega r_*} + B_{\text{out}}e^{+i\omega r_*}$ with $|B_{\text{out}}/B_{\text{in}}| = 1$ exactly for real ω . We write $S(\omega) = B_{\text{out}}/B_{\text{in}} = e^{2i\delta(\omega)}$. The QNMs appear as resonances in $\delta(\omega)$. The same holds for the inviscid star of Boyanov et al. (2024), for which $\mathcal{R}^2 = 1$. To obtain non-trivial reflection and

transmission coefficients we treat the *exterior potential plus the shell* as a scatterer between spatial infinity and the flat interior. The interior is taken to be “open”, i.e. it contains only the ingoing wave $\Psi_- = \hat{h}_\ell^{\text{in}}(\nu r)$, Eq. (3.2). We then define

$$\mathcal{R}(\omega) = \left| \frac{B_{\text{out}}}{B_{\text{in}}} \right|^2, \quad \mathcal{T}(\omega) = \frac{1}{|B_{\text{in}}|^2}. \quad (5.1)$$

Let $J = \text{Im}(\Psi^* \partial_x \Psi)$ be the Wronskian flux in the global coordinate (1.10). It is constant on either side, with $J_{\text{int}} = \sqrt{f_R} \nu \text{Im}(\hat{h}^{\text{in}*} \hat{h}^{\text{in}'}) = -\omega$ and $J_{\text{ext}} = \omega(|B_{\text{out}}|^2 - |B_{\text{in}}|^2)$. If the junction conserves J then $\mathcal{R} + \mathcal{T} = 1$. For real $\mathbf{T}$ this is the statement $\det \mathbf{T} = f_R^{-1/2}$. The identity is *not* built into the construction, so it is a non-trivial test of the junction conditions (Sec. 6.1). The coefficients (5.1) have the limits requested in the task.

- As $\omega \rightarrow 0$, $\mathcal{R} \rightarrow 1$. The irregular static branch $r^{-\ell}$ of the interior Hankel function dominates, and the exterior solution matched to it is a standing wave (the shell’s tidal response).
- As $R \rightarrow 2M$, $\mathcal{R} \rightarrow \mathcal{R}_{\text{BH}}$ (Vishveshwara 1970). The shell sits deep in the near-horizon region, where $V \rightarrow 0$ and $\Delta_{\text{odd}} \rightarrow 0$, so the transmitted wave is exactly the horizon-ingoing one.
- As $M \rightarrow 0$, $\mathcal{R} \rightarrow 0$ (flat space, no scatterer).
- At high frequency, $\mathcal{R} \rightarrow 0$. For odd parity $\mathcal{R} \simeq \Delta_{\text{odd}}^2/(4\omega^2)$, the reflectivity of a δ -barrier. No impedance jump arises because the local wavenumber $\omega/\sqrt{f_R}$ in proper distance is continuous across the shell.

The high-frequency behaviour differs from that of Boyanov et al.’s viscous star, which absorbs at high frequency: the thin shell becomes *transparent*.

5.2 Method 1: transfer matrix with a continued-fraction exterior basis

Let Ψ_{up} be the purely outgoing exterior solution. Following Leins, Nollert & Soffel (1993) and Pani et al. (2009, App. B), we expand it about a point $r = R_2$:

$$\Psi_{\text{up}} = \chi(r) \sum_{n \geq 0} a_n z^n, \quad z = 1 - \frac{R_2}{r}, \quad \chi = (r - 2M)^{2iM\omega} e^{i\omega r}. \quad (5.2)$$

The exact identity $f\chi'/\chi = i\omega$ reduces Eq. (2.2) to $c(z)\phi'' + d(z)\phi' + e(z)\phi = 0$, where $u = 1 - z$, $m = M/R_2$ and $c = u^2(1 - 2mu)$, $d = -2u + 6mu^2 + 2i\omega R_2$, $e = 6mu - \lambda$ (verified in `symbolic/exterior_series.py`). This is Pani et al. Eq. (B4) with $\omega \rightarrow -\omega$. The coefficients obey a four-term recurrence (their Eq. B7) that Gaussian elimination reduces to three terms (their Eqs. B10–B13). The minimal solution gives a_1/a_0 as a continued fraction (their Eq. B17), whose depth is increased adaptively until it converges to 10^{-12} . Then $\Psi'_{\text{up}}/\Psi_{\text{up}}|_{R_2} = i\omega R_2/(R_2 - 2M) + (a_1/a_0)/R_2$. Convergence of (5.2) up to $z \rightarrow 1$ requires the singular point $r = 2M$ to lie outside $|z| < 1$, i.e. $R_2 \geq 4M$. We take $R_2 = \max(R, 4.5M)$ and, when $R < 4.5M$, integrate from R_2 down to R (DOP853). Zerilli solutions follow from the Chandrasekhar map. For real ω , $\Psi_{\text{in}} = \Psi_{\text{up}}^*$, and the normalization $|\Psi_{\text{up}}(R)|^2 = \omega/\text{Im}(\partial_{r_*} \ln \Psi_{\text{up}})$ follows from the Wronskian. The junction data $(\Psi_+, \Psi'_+) = \mathbf{T}(\Psi_-, \Psi'_-)$ then give B_{in} and B_{out} by a 2×2 linear solve.

5.3 Method 2: shooting

We integrate the exterior RW/Zerilli equation in r , from R^+ (junction data) outwards to $r_{\text{far}} = \max(150M, 60/\omega)$. We use `scipy.integrate.solve_ivp` with DOP853, `rtol`= 10^{-10} and `atol`= 10^{-12} , as prescribed. At r_{far} we project onto the asymptotic basis $\Psi_{\text{up}} = e^{i\omega r_*} \sum_k c_k r^{-k}$ and $\Psi_{\text{in}} = \Psi_{\text{up}}^*$. The coefficients obey the recursion

$$2i\omega(k+1)c_{k+1} = [k(k+1) - \lambda]c_k - 2M(k^2 - 4)c_{k-1}, \quad c_0 = 1, \quad (5.3)$$

which follows from Eq. (2.2) with $\Psi = e^{i\omega r_*} u(r)$, $fu'' + (f' + 2i\omega)u' - (\lambda/r^2 - 6M/r^3)u = 0$. The series is truncated at its smallest non-zero term (for $\ell = 2$, $c_3 \equiv 0$). The Zerilli asymptotic basis follows from the Chandrasekhar map. We also evaluate the estimator of Boyanov et al. (2024, App. B1), $|i\omega\Psi + \partial_{r_*}\Psi|/|i\omega\Psi - \partial_{r_*}\Psi| = |B_{\text{out}}/B_{\text{in}}|$, averaged over four periods beyond r_{far} . Methods 1 and 2 share only the junction matrix and the interior functions; their exterior solutions are completely independent. For the closed system, the interior is integrated numerically from $r_0 = 10^{-3}R$ with a power-series start, independently of the Bessel routines.

5.4 QNMs

QNMs are the zeros of the normalized Wronskian at R^+ between the solution coming through the junction from the regular interior, $\mathbf{y}_J = \mathbf{T}(\omega)(\hat{j}_\ell, \nu\hat{j}'_\ell)$, and the outgoing solution: $W = (\Psi_J\Psi'_{\text{up}} - \Psi'_J\Psi_{\text{up}})/(|\mathbf{y}_J||\mathbf{y}_{\text{up}}|)$. The roots are found with `scipy.optimize.root` on $(\text{Re } W, \text{Im } W)$.

- In *Method 1*, Ψ_{up} comes from the continued fraction.
- In *Method 2* we follow the strategy of Boyanov et al. (2024, App. B1): integrate inward from a high-order exterior expansion to the matching point. To make this stable for complex ω we integrate along the complex ray $r = R + se^{i\varphi}$, $\varphi = \pi/2 - \arg \omega$, on which Ψ_{up} is exponentially *subdominant* at large s . We start from the asymptotic series (5.3) at $s = \max(60M, 40/|\omega|)$.

Candidate roots come from a vectorized scan of $\log |W|$ on a grid in the complex plane. An extra logarithmic scan near the real axis is used for compact shells, and post-Newtonian seeds from PSP26 Eq. (7.8) are used for the matter modes. Every root found with Method 1 is re-converged with Method 2.

High-precision arithmetic. The exact even-parity matrices $\mathbf{A}$, $\mathbf{B}$ involve large cancellations near $R = 2M$. In double precision $\det \mathbf{T}$ deviates from $f_R^{-1/2}$ by 10^{-6} at $R = 2.2M$, and the method breaks down for $R \lesssim 2.05M$. For $R < 2.3M$ we therefore evaluate (4.2) with 40-digit `mpmath`. $\det \mathbf{T} \sqrt{f_R} = 1$ is then satisfied to 10^{-15} down to $R = 2.01M$.

5.5 Schwarzschild black hole and WKB

Black hole. We use Leaver’s continued fraction (Leaver 1985) with inversion index equal to the overtone number. The reflectivity with a horizon-ingoing boundary condition is obtained by DOP853 integration from $r = 2M(1 + 10^{-6})$, with a first-order Frobenius start. *WKB.* We implement arbitrary-order WKB through the equivalent perturbative expansion of the inverted anharmonic oscillator (Bender–Wu), computed with symbolic overtone n by the hypervirial–Hellmann–Feynman recursion $2kE\langle x^{k-1} \rangle = 2k\langle x^{k-1}V \rangle + \langle x^k V' \rangle - \frac{1}{2}k(k-1)(k-2)\langle x^{k-3} \rangle$, with the r_* -derivatives of V at its maximum up to 12th order. The terms through $g^{2(N-1)}$ reproduce the N -th order WKB formula: Schutz & Will (1985) for $N = 1$, Iyer & Will (1987) for $N = 3$, and Konoplya (2003) for $N = 6$. We checked this against Iyer & Will’s Table I: at $N = 3$, $\ell = 2$ gives $0.3732 - 0.0892i$ ($n = 0$) and $0.3460 - 0.2749i$ ($n = 1$). For modes trapped between the flat interior and the exterior barrier of ultracompact shells, the WKB estimate is Bohr–Sommerfeld (with Langer’s $\lambda \rightarrow (\ell + \frac{1}{2})^2$) plus Gamow tunnelling: $\text{Im } \omega = -T_b/(4\omega \int dx/\sqrt{\omega^2 - V})$.

6 Results

6.1 Validation checks

All checks requested in the task are collected in Tables 2–4. They are produced by `main.py` `validate` and `make.tables.py`.

Table 2: Validation summary. M1/M2 denote the transfer-matrix and shooting methods. $|\Delta_{12}|$ is the maximum difference between them over the frequency grid $\omega M \in [0.005, 2]$.

Check	value
Odd: generated vs analytic $\mathbf{T}$	5.7e-16
Odd: $\mathbf{T}^A - \mathbf{T}^C$	1.1e-15
Even C: $ \mathbf{T} - \mathbb{I} $ at $M = 10^{-8}$ (transparent limit)	4.0e-09
Even A: $ \mathbf{T} - \mathbb{I} $ at $M = 10^{-8}$	4.7e-09
Even C: $\max \det \mathbf{T} \sqrt{f_R} - 1 $	3.6e-12
Even A: constraint violation, $R = 3M, \omega M = 1$	1.2e-01
Even A: constraint violation, $R = 3M, \omega M = 10$	1.4e-02
Even A: constraint violation, $R = 3M, \omega M = 100$	1.4e-03
Even A: constraint violation, $R = 6M, \omega M = 1$	6.0e-02
Even A: constraint violation, $R = 6M, \omega M = 10$	7.9e-03
Even A: constraint violation, $R = 6M, \omega M = 100$	7.9e-04
odd, $\ell = 2, R = 3M$: $\max_{\omega} \mathcal{R} + \mathcal{T} - 1 $ (M2) / $\max \Delta_{12} $ / Boyanov estimator	5.3e-10 / 5.3e-10 / 2.5e-06
odd, $\ell = 2, R = 6M$: $\max_{\omega} \mathcal{R} + \mathcal{T} - 1 $ (M2) / $\max \Delta_{12} $ / Boyanov estimator	6.0e-10 / 6.0e-10 / 2.0e-06
even-C, $\ell = 2, R = 3M$: $\max_{\omega} \mathcal{R} + \mathcal{T} - 1 $ (M2) / $\max \Delta_{12} $ / Boyanov estimator	5.3e-10 / 5.3e-10 / 4.0e-06
even-C, $\ell = 2, R = 6M$: $\max_{\omega} \mathcal{R} + \mathcal{T} - 1 $ (M2) / $\max \Delta_{12} $ / Boyanov estimator	6.1e-10 / 6.1e-10 / 3.2e-06
even-A, $\ell = 2, R = 3M$: $\max_{\omega} \mathcal{R} + \mathcal{T} - 1 $ (M2) / $\max \Delta_{12} $ / Boyanov estimator	5.4e-01 / 6.0e-10 / 2.5e-06
even-A, $\ell = 2, R = 6M$: $\max_{\omega} \mathcal{R} + \mathcal{T} - 1 $ (M2) / $\max \Delta_{12} $ / Boyanov estimator	2.2e+00 / 6.1e-10 / 3.6e-06
Low frequency: $\max 1 - \mathcal{R} $ at $\omega M = 10^{-3}$ (all models)	8.9e-16

Energy conservation. For odd parity and for model C in the even sector, $|\mathcal{R} + \mathcal{T} - 1| \lesssim 10^{-10}$ over the whole range $\omega M \in [0.005, 2]$ (Fig. 6). This is better than the required 10^{-6} by four orders of magnitude, with $\mathcal{R}$ and $\mathcal{T}$ obtained independently from the asymptotic amplitudes (Method 2). Since the identity is not imposed anywhere, it is a stringent test of the even-parity junction conditions. For model A the violation is physical: it measures the error of the frozen approximation and scales as ω^{-2} (Fig. 6, right).

Consistency of the two methods. Transfer matrix and shooting agree on $\mathcal{R}$ and $\mathcal{T}$ to 10^{-10} (Fig. 5), and on QNM frequencies to 10^{-8} – 10^{-13} (Tables 10 and 6). The required tolerance is 10^{-4} . The windowed estimator of Boyanov et al. (2024) agrees with the projected amplitudes to the accuracy expected from its neglect of $O(1/r)$ corrections at r_{far} .

Pure Schwarzschild limit. Our exterior machinery reproduces the Schwarzschild QNMs (Table 3). For $\ell = 2, n = 0$ we find $\omega M = 0.37367168 - 0.08896232i$, in agreement with the literature value $0.3736716844 - 0.0889623157i$ (Leaver 1985; Berti, Cardoso & Starinets 2009) to all digits shown. All overtones $n \leq 3$ for $\ell = 2, 3, 4$ agree with the standard tables to six digits. The black-hole reflectivity (horizon-ingoing condition) is identical for the RW and Zerilli potentials to 10^{-7} , as required by isospectrality. It reproduces Vishveshwara (1970) Fig. 2: $\mathcal{R} = 1.55 \times 10^{-2}$ at $(2M\omega)^2 = 1$ and 5.3×10^{-9} at $(2M\omega)^2 = 4$, consistent with the log-plot. For the shell, the transparent limit $M \rightarrow 0$ gives $\mathbf{T} \rightarrow \mathbb{I}$. As $R \rightarrow 2M$, $\mathcal{R}(\omega) \rightarrow \mathcal{R}_{\text{BH}}(\omega)$ (Table 4, Fig. 7). The QNM spectrum, however, does *not* approach the black-hole spectrum. As $R \rightarrow 2M$ the modes become trapped between the interior and the exterior barrier, and they approach the real axis. This is the behaviour of all horizonless ultracompact objects (cf. Pani et al. 2009; Cardoso, Franzin & Pani 2016).

Low frequency. $1 - \mathcal{R} < 10^{-11}$ at $\omega M = 10^{-3}$ for all models: the shell is opaque to long-wavelength waves, as the star of Boyanov et al. (2024, Fig. 1) is.

WKB vs numerical. Table 3 and Fig. 12 compare WKB orders 1, 3, 6 with Leaver’s values. The errors follow the known pattern. For $\ell = 2, n = 0$ they are 7×10^{-2} , 1.5×10^{-3} and 2.3×10^{-4} .

The accuracy improves with ℓ and degrades with n . For the trapped modes of ultracompact shells (Table 9), Bohr–Sommerfeld/Gamow WKB gives $\text{Re } \omega$ to 1–6% and $\text{Im } \omega$ to within a factor $\lesssim 2$. That is the expected accuracy for low-lying modes: WKB neglects the shell coupling and the Langer-corrected centrifugal region is short.

Table 3: Schwarzschild QNMs ωM : Leaver continued fraction (this code) vs WKB of orders 1, 3, 6 (hypervirial construction). Literature: Leaver (1985); Berti, Cardoso & Starinets (2009); Iyer & Will (1987) Table I for 3rd order; Konoplya (2003) for 6th order.

ℓ	n	Leaver CF (this code)	WKB 1st	WKB 3rd	WKB 6th
2	0	0.37367168 – 0.08896232 <i>i</i>	0.3988 – 0.0883 <i>i</i>	0.37316 – 0.08922 <i>i</i>	0.373619 – 0.088891 <i>i</i>
2	1	0.34671100 – 0.27391488 <i>i</i>	0.4534 – 0.2330 <i>i</i>	0.34602 – 0.27492 <i>i</i>	0.346297 – 0.273480 <i>i</i>
2	2	0.30105345 – 0.47827698 <i>i</i>	0.5170 – 0.3406 <i>i</i>	0.30293 – 0.47106 <i>i</i>	0.298520 – 0.477560 <i>i</i>
2	3	0.25150496 – 0.70514820 <i>i</i>	0.5775 – 0.4268 <i>i</i>	0.24746 – 0.67290 <i>i</i>	0.241715 – 0.709595 <i>i</i>
3	0	0.59944329 – 0.09270305 <i>i</i>	0.6166 – 0.0923 <i>i</i>	0.59927 – 0.09273 <i>i</i>	0.599443 – 0.092703 <i>i</i>
3	1	0.58264380 – 0.28129811 <i>i</i>	0.6619 – 0.2580 <i>i</i>	0.58235 – 0.28141 <i>i</i>	0.582642 – 0.281290 <i>i</i>
3	2	0.55168490 – 0.47909275 <i>i</i>	0.7251 – 0.3925 <i>i</i>	0.55320 – 0.47668 <i>i</i>	0.551594 – 0.479047 <i>i</i>
3	3	0.51196191 – 0.69033710 <i>i</i>	0.7909 – 0.5038 <i>i</i>	0.51575 – 0.67743 <i>i</i>	0.511099 – 0.690468 <i>i</i>
4	0	0.80917838 – 0.09416396 <i>i</i>	0.8223 – 0.0939 <i>i</i>	0.80910 – 0.09417 <i>i</i>	0.809178 – 0.094164 <i>i</i>
4	1	0.79663153 – 0.28433435 <i>i</i>	0.8602 – 0.2694 <i>i</i>	0.79650 – 0.28437 <i>i</i>	0.796631 – 0.284334 <i>i</i>
4	2	0.77270953 – 0.47990818 <i>i</i>	0.9187 – 0.4204 <i>i</i>	0.77364 – 0.47897 <i>i</i>	0.772695 – 0.479900 <i>i</i>
4	3	0.73983673 – 0.68392432 <i>i</i>	0.9844 – 0.5492 <i>i</i>	0.74331 – 0.67830 <i>i</i>	0.739665 – 0.683902 <i>i</i>

Table 4: Approach to the black-hole reflectivity as $R \rightarrow 2M$ ($\ell = 2$; even parity is model C, $\Gamma = 2$, evaluated in 40-digit arithmetic).

ωM	BH	odd 2.2 <i>M</i>	odd 2.05 <i>M</i>	odd 2.002 <i>M</i>	even-C 2.2 <i>M</i>	even-C 2.05 <i>M</i>	even-C 2.01 <i>M</i>
0.20	0.99913	0.99886	0.99916	0.99907	0.99966	0.99921	0.99915
0.30	0.94521	0.92247	0.93661	0.94729	0.96577	0.95037	0.94996
0.37	0.56378	0.46896	0.52601	0.56105	0.62593	0.61107	0.54639
0.45	0.07559	0.04894	0.10167	0.07539	0.13163	0.07064	0.07361
0.60	0.00066	0.00554	0.00319	0.00021	0.00814	0.00166	0.00141

6.2 Scattering coefficients

Figures 2–3 show $\mathcal{R}(\omega)$ and $\mathcal{T}(\omega)$ for both parities and both models.

1. At low frequency the shell is a perfect reflector. The transition to transparency occurs at $\omega \sim \sqrt{V_{\text{max}}}$ for compact shells ($R \lesssim 3M$), where the exterior potential barrier dominates. For dilute shells it occurs at $\omega R \sim \ell$, where the flat-space centrifugal barrier is crossed.
2. For compact shells the curves approach the Schwarzschild reflectivity. The even and odd curves differ, because the even junction is frequency dependent. The shell spacetime breaks the isospectrality of Schwarzschild, as PSP26 also note for the QNMs.
3. At high frequency $\mathcal{R} \rightarrow 0$, with the δ -barrier law $\mathcal{R} \simeq \Delta_{\text{odd}}^2/4\omega^2$ in the odd sector. The thin shell becomes transparent. This differs from the viscous star of Boyanov et al., which absorbs.
4. Even parity, model C, shows resonant features at the matter-mode frequencies $\omega \sim (M/R^3)^{1/2}$. Their EOS dependence is shown in Fig. 4. Away from them, the dependence on Γ is weak, in line with PSP26’s observation that even wave modes are almost independent of Γ .
5. Model C (even) merges with the odd-parity curve at high frequency: both follow the δ -barrier law, because the fluid shell’s inertia and pressure response become irrelevant. Model A does

not: the frozen domain wall reflects much more strongly, and its $\mathcal{R}$ decays more slowly with ω (Figs. 2, 11). The wall has no compressional degree of freedom. Its only even response is the displacement of a membrane under tension σ , which couples to the wave through the tension terms of $\delta S_{ab} = -\sigma \delta h_{ab}$. At low frequency ($\omega \lesssim \omega_{\text{dyn}}$) model A also violates energy conservation at $O(1)$. This is the breakdown of the adiabatic approximation, not a numerical error.

6. At the single point $\ell = 4$, $\omega M = 0.0057$, Method 1 returns no value: $\mathcal{T} \sim 10^{-25}$ underflows in the Wronskian normalization. The point is omitted from the plots; at neighbouring frequencies $\mathcal{R} = 1$ to machine precision.

6.3 Quasinormal modes

The QNM spectra are shown in Fig. 8 (complex plane, shell vs Schwarzschild) and Fig. 9 (dependence on R/M). Selected values are listed in Tables 6–8. The complete list is in `results/qnm.table.csv`.

- *Wave modes* exist in both sectors, with $\omega \sim 1/R$. For dilute shells they follow PSP26’s Newtonian law $\text{Re}(\omega R) \simeq n\pi$ or $(n + \frac{1}{2})\pi$, $\text{Im}(\omega R) \simeq -\xi$ with $\xi = \frac{1}{2} \ln[2\ell(\ell+1)R/M]$ (PSP26 Eqs. 7.23–7.25). The imaginary part grows only logarithmically with R/M : the shell is a weak reflector, and the interior cavity leaks. Quantitatively, our wave modes coincide with PSP26’s exact numerical results (their Figs. 3–4) to the precision with which the figures can be read (± 0.02); see Table 5.
- For compact shells ($R < 3M$) a family of *trapped* modes appears below the barrier top. Their damping is set by tunnelling, and as $R \rightarrow 2M$ they become exponentially long-lived and dense (Table 9). They are the modes responsible for GW “echoes”. None of them tends to the Schwarzschild QNMs.
- *Matter modes* (even parity, model C only) are discussed next.

Table 5: Wave modes ($\ell = 2$; even sector: model C, $\Gamma = 2$) compared with the values read off PSP26 Figs. 3 and 4 (reading uncertainty $\approx \pm 0.02$); $\xi = \frac{1}{2} \ln[2\ell(\ell+1)R/M]$.

parity	M/R	mode	$\text{Re}(R\omega)$	$\text{Im}(R\omega)/\xi$	PSP26 Re	PSP26 Im/ ξ
odd	0.167	1	1.484	-0.632	1.47	-0.630
odd	0.167	2	4.151	-0.835	4.10	-0.840
even	0.250	1	1.033	-0.651	1.05	-0.655
even	0.100	1	0.615	-0.825	0.62	-0.825
even	0.100	2	2.978	-0.907	2.95	-0.910

6.4 Even-parity matter modes and the instability of the static shell

The literature search (Sec. 8) found that our system, the static perfect-fluid shell with Minkowski interior and Schwarzschild exterior, has been studied very recently by Pitre, Schneider & Poisson [PSP26, arXiv:2604.05980]. They found four even-parity “matter modes” per ℓ : an unstable purely imaginary one, its damped mirror, and a pair $\pm \text{Re} \omega + i \text{Im} \omega$ with small negative imaginary part. They concluded that the shell is dynamically unstable on all angular scales. Their method is different from ours: retarded-time slicing, an RW-type master function in both sectors, and Chebyshev collocation / MST. We therefore reproduce their results as an independent validation of our even-parity junction conditions (Table 10, Fig. 10).

- At $M/R = 0.2$, $\Gamma = 2$, $\ell = 2$ we find $\omega(R^3/M)^{1/2} = 0.60771 i$ for the unstable mode and $1.26707 - 0.00110 i$ for the stable pair. These agree with PSP26 Figs. 1 and 2, which show $\approx 0.61 i$ and $\approx 1.27 - 1.1 \times 10^{-3} i$.

Table 6: Lowest wave modes, $\ell = 2$.

R/M	sector	lowest wave modes ωM (ordered by $\text{Re } \omega$; Methods 1 and 2 agree to $< 10^{-4}$)
2.2	odd	$0.42111 - 0.03502i, 0.66196 - 0.19736i, 0.66692 - 0.28043i, 0.78377 - 0.71437i$
2.2	even-C	$0.44694 - 0.04385i, 0.57996 - 0.19970i, 0.69756 - 0.56050i, 0.92055 - 0.21412i$; matter: $0.00000 + 0.24429i, 0.00000 - 0.24429i$
2.2	even-A	$0.10284 - 0.04070i, 0.45445 - 0.10428i, 0.56024 - 0.44388i, 0.65595 - 0.85878i$
3	odd	$0.20371 - 1.01489i, 0.35352 - 0.55904i, 0.42638 - 0.17728i, 0.97955 - 0.33508i$
3	even-C	$0.21102 - 0.76599i, 0.40766 - 0.29425i, 0.61594 - 0.31862i, 1.91546 - 0.38053i$; matter: $-0.00000 + 0.13321i, -0.00000 - 0.13326i$
3	even-A	$0.05288 - 0.97817i, 0.24913 - 0.59916i, 0.35675 - 0.22995i, 0.96224 - 0.13441i$
6	odd	$0.24739 - 0.22534i, 0.69189 - 0.29749i, 1.13482 - 0.31819i, 1.57166 - 0.33553i$
6	even-C	$0.13710 - 0.26885i, 0.45011 - 0.28793i, 0.91288 - 0.30736i, 1.35361 - 0.32702i$; matter: $0.08942 - 0.00005i, -0.00000 + 0.04014i, -0.00000 - 0.04014i$
6	even-A	$0.18120 - 0.22393i, 0.66401 - 0.17021i, 1.12337 - 0.19838i, 1.56453 - 0.21844i$
10	odd	$0.15342 - 0.18490i, 0.45405 - 0.22153i, 0.74518 - 0.23457i, 1.03239 - 0.24567i$
10	even-C	$0.06152 - 0.19748i, 0.29784 - 0.21714i, 0.59930 - 0.22783i, 0.88902 - 0.24022i$; matter: $0.04421 - 0.00001i, -0.00000 + 0.01762i, 0.00000 - 0.01762i$
10	even-A	$0.11383 - 0.17403i, 0.43293 - 0.14212i, 0.73603 - 0.15791i, 1.02659 - 0.17041i$

Table 7: Lowest wave modes, $\ell = 3$.

R/M	sector	lowest wave modes ωM (ordered by $\text{Re } \omega$; Methods 1 and 2 agree to $< 10^{-4}$)
2.2	odd	$0.61197 - 0.01610i, 0.79559 - 0.14940i, 0.92563 - 0.26191i, 1.32591 - 0.24723i$
2.2	even-C	$0.62406 - 0.01797i, 0.94668 - 0.42655i, 1.05770 - 0.83502i, 1.09205 - 0.23717i$; matter: $0.00000 + 0.33715i, 0.00000 - 0.33715i$
2.2	even-A	$0.66172 - 0.05295i, 0.80201 - 0.33708i, 0.86783 - 0.05466i, 0.90221 - 0.70492i$
3	odd	$0.45895 - 0.96508i, 0.60569 - 0.55699i, 0.65509 - 0.18820i, 1.22298 - 0.38277i$
3	even-C	$0.35057 - 1.17392i, 0.54273 - 0.73309i, 0.66906 - 0.23954i, 0.83640 - 0.42456i$; matter: $0.31096 - 0.00003i, 0.00000 + 0.22263i, -0.00000 - 0.22263i$
3	even-A	$0.36818 - 0.96443i, 0.52224 - 0.58563i, 0.61501 - 0.22237i, 1.18558 - 0.16840i$
6	odd	$0.19277 - 0.46109i, 0.40294 - 0.26528i, 0.86444 - 0.33236i, 1.31485 - 0.33882i$
6	even-C	$0.05982 - 0.50984i, 0.33182 - 0.32097i, 0.59758 - 0.34764i, 1.08605 - 0.33206i$; matter: $0.13334 - 0.00000i, 0.00000 + 0.07705i, -0.00000 - 0.07705i$
6	even-A	$0.15669 - 0.44150i, 0.35437 - 0.25459i, 0.82650 - 0.19835i, 1.30131 - 0.21402i$
10	odd	$0.07849 - 0.31820i, 0.25779 - 0.21802i, 0.56823 - 0.24477i, 0.86393 - 0.24851i$
10	even-C	$0.18733 - 0.24644i, 0.39972 - 0.25566i, 0.71377 - 0.24471i, 1.00967 - 0.25121i$; matter: $0.06489 - 0.00000i, -0.00000 + 0.03530i, 0.00000 - 0.03530i$
10	even-A	$0.06376 - 0.29784i, 0.22599 - 0.20104i, 0.54149 - 0.16173i, 0.85350 - 0.16877i$

- As $M/R \rightarrow 0$ our values approach the post-Newtonian limits of PSP26 Eq. (7.8), 0.51470 i and 1.50496. At $M/R = 0.01$ the differences are 0.0040 and 0.0098, of the size of the expected $O(M/R)$ correction.

This settles the question raised in Sec. 1.3. The truly static model C is linearly unstable, with growth time $\sim 1.6(R^3/M)^{1/2}$. This is comparable to the collapse time of the domain wall of model A. Neither background is stable on the dynamical timescale. The real-frequency scattering results are valid in the high-frequency regime stated at the beginning of this document.

7 Discussion

1. **Shell model.** No static configuration exists for a pure domain wall or a dust shell. A static shell needs positive surface pressure, $p/\sigma = (1 - \sqrt{f_R})/(4\sqrt{f_R})$. Even that shell is dynamically unstable (PSP26, confirmed here). The “collapse problem” of the prior notes is therefore generic: every thin shell of this type evolves on the timescale $(R^3/M)^{1/2}$. GW scattering in the frequency domain is an adiabatic concept, valid for $\omega(R^3/M)^{1/2} \gg 1$. We quantified the accuracy of the adiabatic domain-wall model, and found it to be $O(\omega_{\text{dyn}}/\omega)$.
2. **Junction conditions.** Odd parity: $[\Psi] = 0$ and $[\partial_x \Psi] = \Delta_{\text{odd}} \Psi$, with $\Delta_{\text{odd}} = \sqrt{f_R}(1 - \sqrt{f_R})/R$, independent of ω and of the equation of state. The $[K]^{(0)}$ term that worried the prior notes cancels against δS_{TA} . Even parity: a frequency-dependent 2×2 matrix. The radial

Table 8: Lowest wave modes, $\ell = 4$.

R/M	sector	lowest wave modes ωM (ordered by $\text{Re } \omega$; Methods 1 and 2 agree to $< 10^{-4}$)
2.2	odd	$0.78350 - 0.00632i, 0.96055 - 0.10943i, 1.12058 - 0.23837i, 1.21974 - 0.52330i$
2.2	even-C	$0.79313 - 0.00713i, 0.95445 - 0.10853i, 1.25457 - 0.26967i, 1.72821 - 0.26728i$; matter: $1.27806 - 0.72384i$
2.2	even-A	$0.84021 - 0.02540i, 1.01805 - 0.27016i, 1.02364 - 0.06398i, 1.95769 - 0.11146i$
3	odd	$0.83612 - 0.55496i, 0.86734 - 0.19128i, 1.46401 - 0.42352i, 2.10831 - 0.42242i$
3	even-C	$0.62773 - 1.12565i, 0.81795 - 0.71396i, 0.87869 - 0.22137i, 1.05472 - 0.49667i$; matter: $0.42113 - 0.00000i, 0.00000 + 0.30830i, 0.00000 - 0.30830i$
3	even-A	$0.76236 - 0.57809i, 0.83678 - 0.22130i, 1.40641 - 0.19861i, 2.09450 - 0.22031i$
6	odd	$0.11490 - 0.66203i, 0.34725 - 0.50399i, 0.55217 - 0.29359i, 1.03466 - 0.36255i$
6	even-C	$0.24097 - 0.55312i, 0.50342 - 0.35230i, 0.74190 - 0.40586i, 1.25433 - 0.35402i$; matter: $0.17772 - 0.00000i, -0.00000 + 0.11153i, -0.00000 - 0.11153i$
6	even-A	$0.08922 - 0.62756i, 0.30788 - 0.47992i, 0.50763 - 0.28112i, 0.98618 - 0.22296i$
10	odd	$0.18066 - 0.35695i, 0.35821 - 0.24238i, 0.68050 - 0.26493i, 0.97986 - 0.26164i$
10	even-C	$0.09507 - 0.37186i, 0.30022 - 0.27956i, 0.49865 - 0.29120i, 0.82490 - 0.25981i$; matter: $0.08609 - 0.00000i, -0.00000 + 0.05180i, -0.00000 - 0.05180i$
10	even-A	$0.16134 - 0.33379i, 0.32745 - 0.22396i, 0.64779 - 0.17880i, 0.96788 - 0.17874i$

 Table 9: Trapped modes of ultracompact shells (odd, $\ell = 2$): exact vs WKB.

R/M	exact (M1)	WKB (Bohr–Sommerfeld + Gamow)
2.01	$0.162603 - 9.711e - 07 i$	$0.152432 - 5.081e - 07 i$
2.01	$0.248683 - 5.284e - 05 i$	$0.246194 - 3.249e - 05 i$
2.01	$0.326133 - 8.798e - 04 i$	$0.328792 - 7.287e - 04 i$
2.02	$0.216610 - 1.809e - 05 i$	$0.205905 - 8.922e - 06 i$
2.02	$0.322125 - 1.072e - 03 i$	$0.324183 - 8.289e - 04 i$
2.05	$0.302815 - 8.384e - 04 i$	$0.296095 - 4.811e - 04 i$
2.10	$0.367989 - 8.195e - 03 i$	$0.370495 - 7.304e - 03 i$

displacement of the shell differs between the two RW gauges, and the shell’s compressional dynamics enter through v_s^2 .

3. **Relation to previous work.** Our odd junction agrees with PSP26 and with Pani et al. (2009). Our even junction reproduces PSP26’s matter modes. Pani et al.’s gravastar has $\Sigma = 0$ and a de Sitter interior, so no direct numerical comparison is possible; we did check agreement of the structural matching conditions. PSP26 argue that unstable modes should also exist in Pani et al.’s model.

4. **What is new here** relative to PSP26:

- the reflection and transmission coefficients $\mathcal{R}$ and $\mathcal{T}$, with their low-frequency, high-frequency and black-hole limits;
- the explicit δ -function form of the odd junction;
- the adiabatic domain-wall model and the quantification of its error;
- two independent numerical methods agreeing to $\lesssim 10^{-8}$.

5. **Limitations.**

- Model C is restricted to $R \geq 2.1M$ by the energy and causality conditions.
- The frozen-wall QNMs are meaningful only for $|\omega| \gg \omega_{\text{dyn}}$.
- Like PSP26, we do not attempt a time-domain evolution of the unstable shell.

8 References and equation map

[RW57] T. Regge and J. A. Wheeler, *Stability of a Schwarzschild singularity*, Phys. Rev. **108**, 1063 (1957).
Used: Eq. (25) (odd potential), Eqs. (18), (20) (RW gauge).

Table 10: Even-parity matter modes, model C, $\ell = 2$, $\Gamma = 2$. $|\Delta_{12}|$ is the difference between Methods 1 and 2. Compare PSP26 Figs. 1–2 and their Eq. (7.8).

M/R	unstable: $\omega(R^3/M)^{1/2}$	$ \Delta_{12} $	stable pair: $\omega(R^3/M)^{1/2}$	$ \Delta_{12} $
0.30	0.668693 i	1.6e-12	1.095903 – 1.70e – 03 i	1.8e-12
0.25	0.636634 i	7.1e-12	1.188088 – 1.51e – 03 i	9.2e-13
0.20	0.607709 i	1.6e-11	1.267067 – 1.10e – 03 i	3.9e-13
0.15	0.581398 i	2.3e-11	1.336362 – 6.47e – 04 i	5.1e-13
0.10	0.557313 i	3.3e-11	1.398197 – 2.78e – 04 i	6.1e-13
0.05	0.535158 i	4.6e-11	1.454057 – 5.79e – 05 i	7.4e-13
0.02	0.522690 i	4.6e-11	1.485144 – 6.52e – 06 i	8.3e-13
0.01	0.518662 i	6.1e-11	1.495140 – 1.20e – 06 i	0.0e+00
$\rightarrow 0$ (PN, PSP26 Eq. 7.8)	0.514695 i		1.504962	

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- [PSP26] T. Pitre, B. Schneider and E. Poisson, *Self-gravitating thin shells are dynamically unstable on all angular scales*, arXiv:2604.05980 [gr-qc] (2026). **Same physical system as our model C.** Used: Eqs. (1.1)–(1.2), (2.4)–(2.6) (polytropic shell, Γ), (5.23)/(97) (odd junction: agreement), (7.8) (post-Newtonian matter modes), (7.23)–(7.25) (Newtonian wave modes), Figs. 1–4 (quantitative comparison).
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Honesty statement. A literature search (arXiv/Google Scholar, queries “thin shell quasinormal modes Minkowski interior”, “thin-shell gravastar perturbations”) found that the QNM spectrum of our model C has been computed by PSP26. We reproduce their matter and wave modes and cite them as a validation, not as a new result. We found no published reflection/transmission coefficients for this system.

Corrections to the starting documents and to the task statement.

1. The regular interior solution is $\hat{j}_\ell(\nu r)$ with $\nu = \omega/\sqrt{f_R}$, not $\hat{j}_\ell(\omega r)$, when ω is the exterior Killing frequency (Sec. 1.4).
2. A static dust shell (“Option B”) does not exist. The formula $M = 4\pi\sigma R^2\sqrt{f}$ is incorrect (Sec. 1.3).
3. The background-trace term $[K]^{(0)}$ of `burbuja_matching_israel` Eq. (13) cancels against the pressure term of $\delta S_{\tau A}$ (Sec. 4.2). The $\ddot{R}$ dependence therefore drops out of the odd sector exactly. In the even sector it survives, but model A must then be treated as an $O(\omega_{\text{dyn}}/\omega)$ approximation.
4. Pure-Schwarzschild QNMs are not a limit of the shell spectrum. The $\ell = 2, n = 0$ value is reproduced by our exterior solver with black-hole boundary conditions, and $\mathcal{R}(\omega)$ tends to the black-hole value as $R \rightarrow 2M$, but the shell QNMs do not (Sec. 6.1).

A Code map

<code>code/main.py</code>	<code>driver: validate, scatter, qnm, tracks, csv, figures</code>
<code>code/symbolic/vacuum_relations.py</code>	Einstein-tensor linearization; Eqs. (2.3), (2.6)–(2.8)
<code>code/symbolic/derive_junction.py</code>	first-principles junction conditions; exports A, B
<code>code/symbolic/exterior_series.py</code>	continued-fraction recurrence; Chandrasekhar map
<code>code/shellgw/background.py</code>	shell models (Sec. 1)
<code>code/shellgw/interior.py,</code> <code>exterior.py</code>	interior functions; exterior ODE, asymptotic series, continued fraction
<code>code/shellgw/junction.py</code>	transfer matrices (odd analytic; even generated; mpmath near $2M$)
<code>code/shellgw/scattering.py, qnm.py</code>	Methods 1 and 2 for $\mathcal{R}, \mathcal{T}$ and QNMs
<code>code/shellgw/blackhole.py, wkb.py</code>	Leaver; arbitrary-order WKB; trapped-mode WKB
<code>results/</code>	JSON results, <code>qnm_table.csv</code> , logs
<code>figures/</code>	all figures (PDF and PNG)

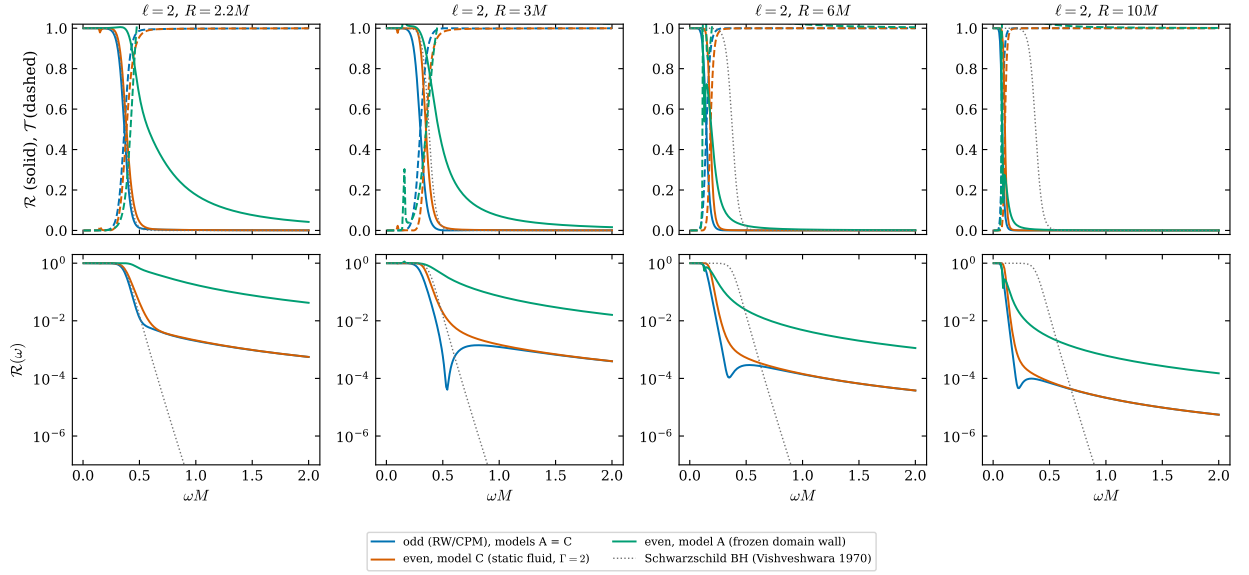


Figure 2: Reflection ($\mathcal{R}$, solid) and transmission ($\mathcal{T}$, dashed; top row) coefficients, Eq. (5.1), for $\ell = 2$ and $R/M = 2.2, 3, 6, 10$. Bottom row: $\mathcal{R}$ on a log scale. Blue: odd parity (models A and C coincide). Orange: even parity, model C ($\Gamma = 2$). Green: even parity, model A. Grey dotted: Schwarzschild black hole.

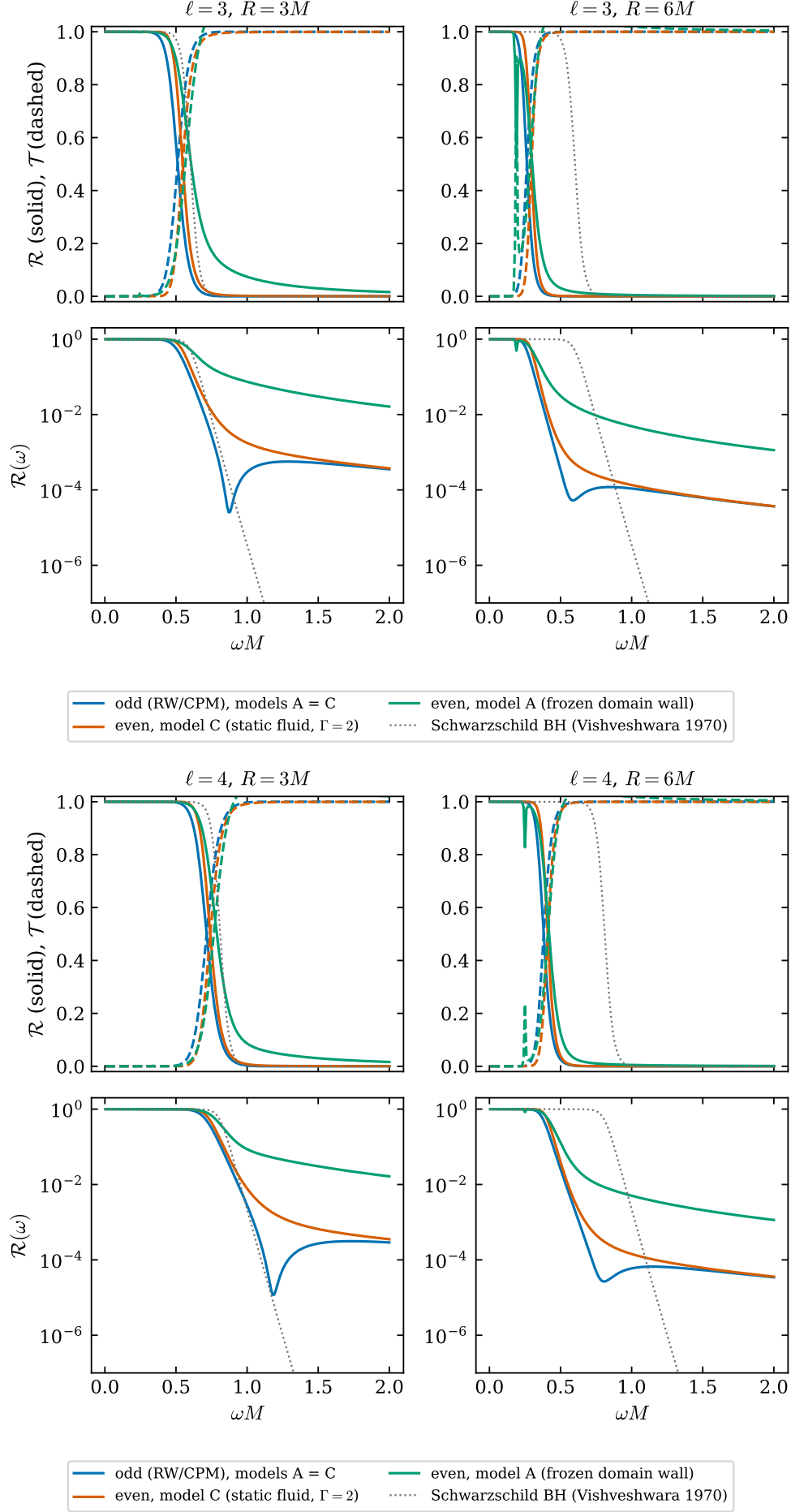


Figure 3: As Fig. 2 for $\ell = 3$ (top) and $\ell = 4$ (bottom), $R/M = 3, 6$.

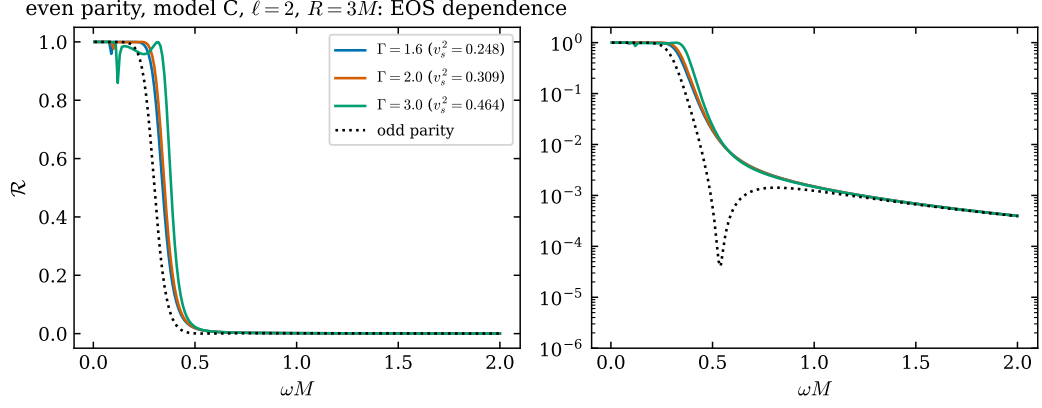


Figure 4: Dependence of the even-parity reflectivity (model C, $\ell = 2$, $R = 3M$) on the adiabatic index Γ of the shell matter.

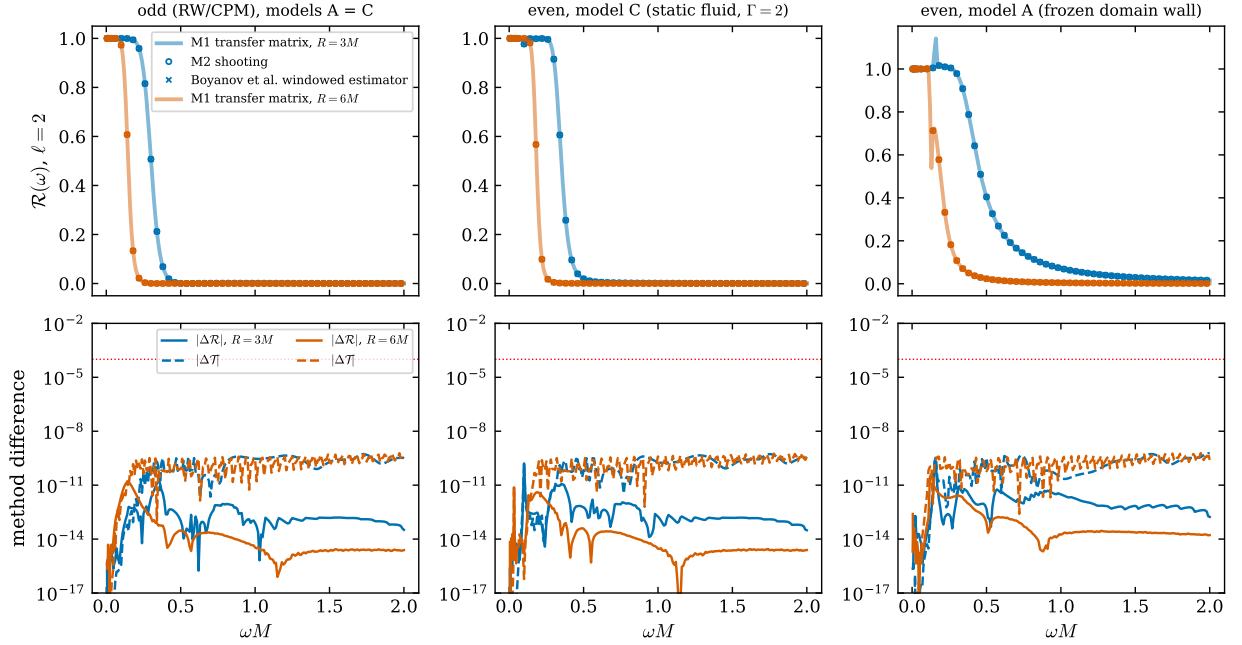


Figure 5: Transfer matrix (Method 1, lines) vs shooting (Method 2, circles) and the windowed estimator of Boyanov et al. (crosses). The bottom row shows the absolute differences; the red line marks the required tolerance 10^{-4} .

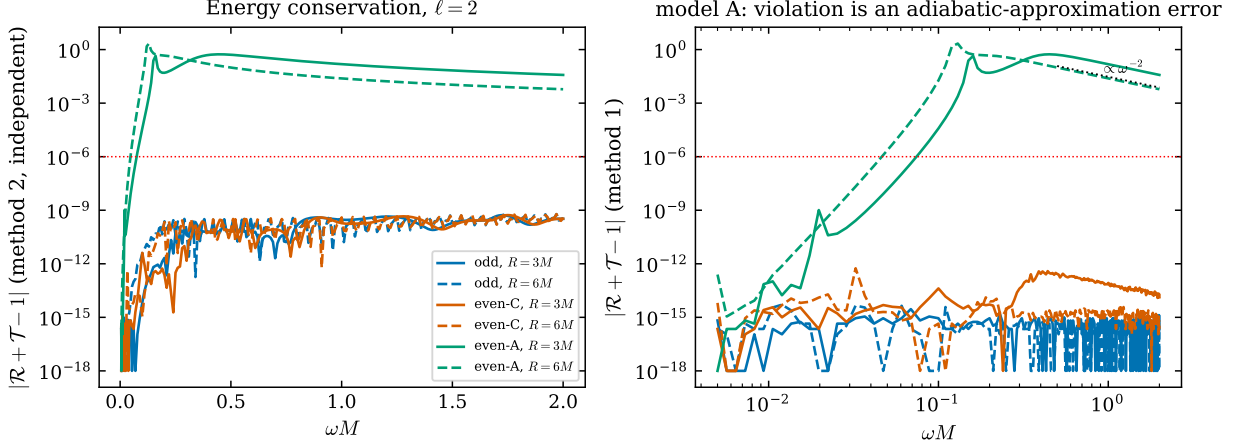


Figure 6: Energy conservation $|\mathcal{R} + \mathcal{T} - 1|$. Left: Method 2, in which $\mathcal{R}$ and $\mathcal{T}$ are obtained independently from asymptotic amplitudes. Right: Method 1 on log-log axes, showing that the model-A violation decays as ω^{-2} (adiabatic error). The red line marks 10^{-6} .

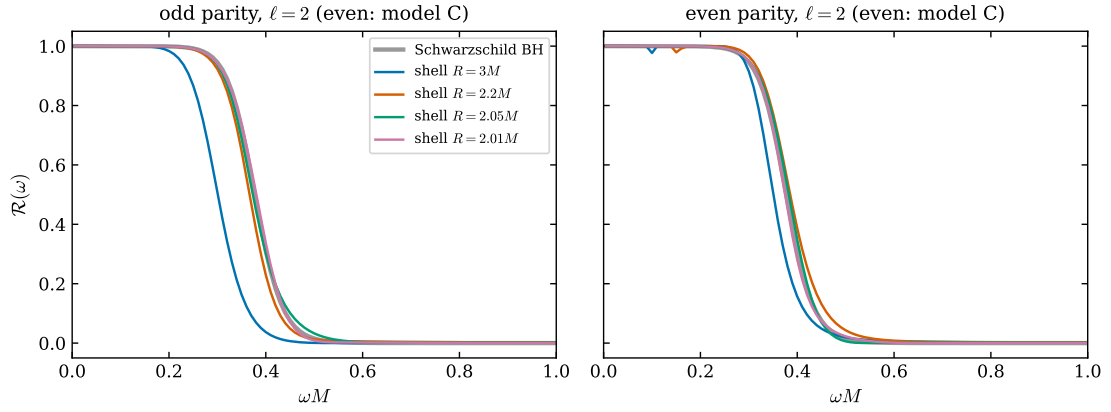


Figure 7: Approach of the shell reflectivity to the Schwarzschild black-hole value (Vishveshwara 1970) as $R \rightarrow 2M$.

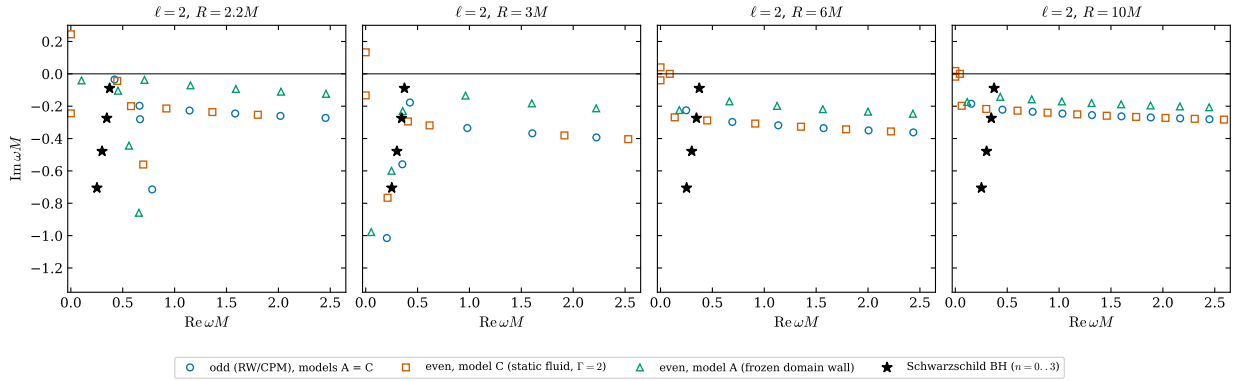


Figure 8: QNMs in the complex ω plane for $\ell = 2$: odd sector (circles), even sector model C (squares), even sector model A (triangles), and Schwarzschild QNMs (stars). The model-C unstable matter mode lies on the positive imaginary axis, at $\omega M \approx +i 0.6(M/R)^{3/2}$.

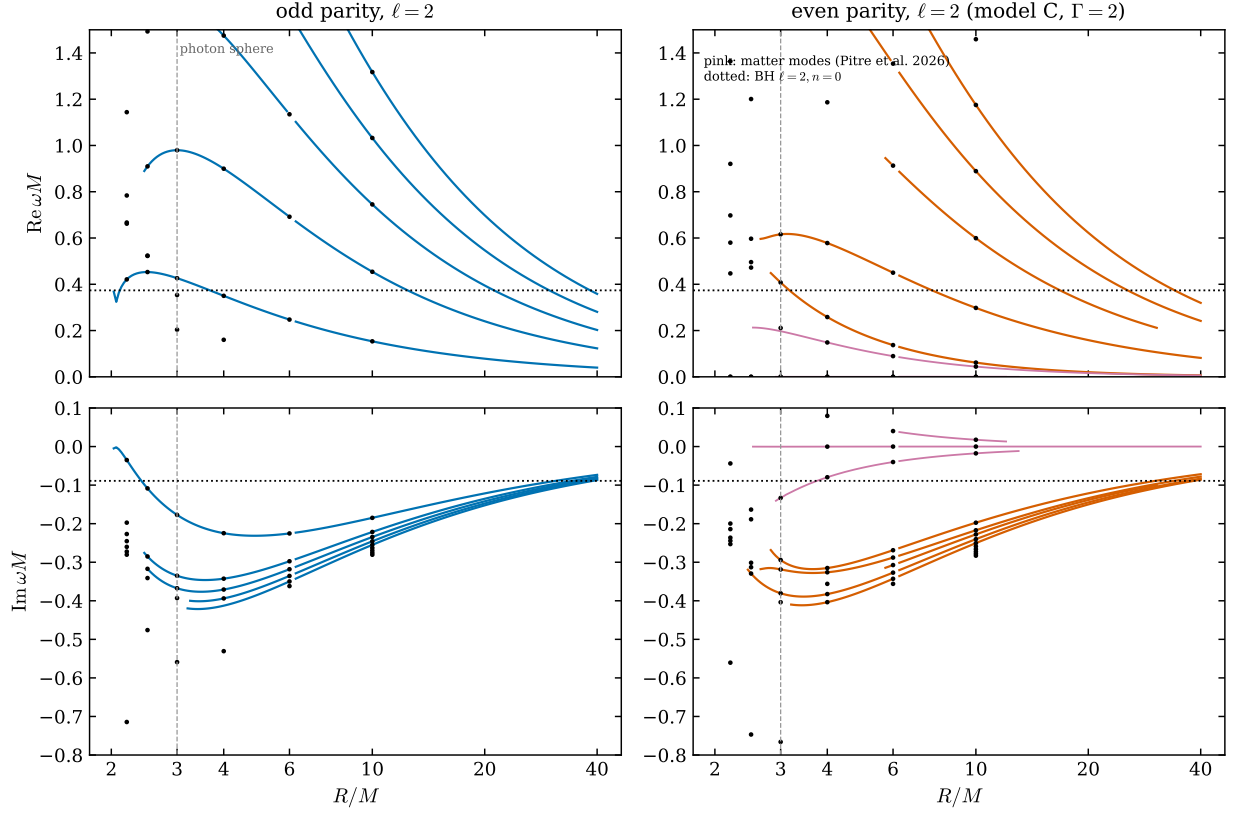


Figure 9: QNM frequencies vs R/M for $\ell = 2$ (continuation in R from the $R = 6M$ catalogue; dots are the independent scans at fixed R). The dotted line marks the Schwarzschild $n = 0$ mode and the dashed line the photon sphere $R = 3M$.

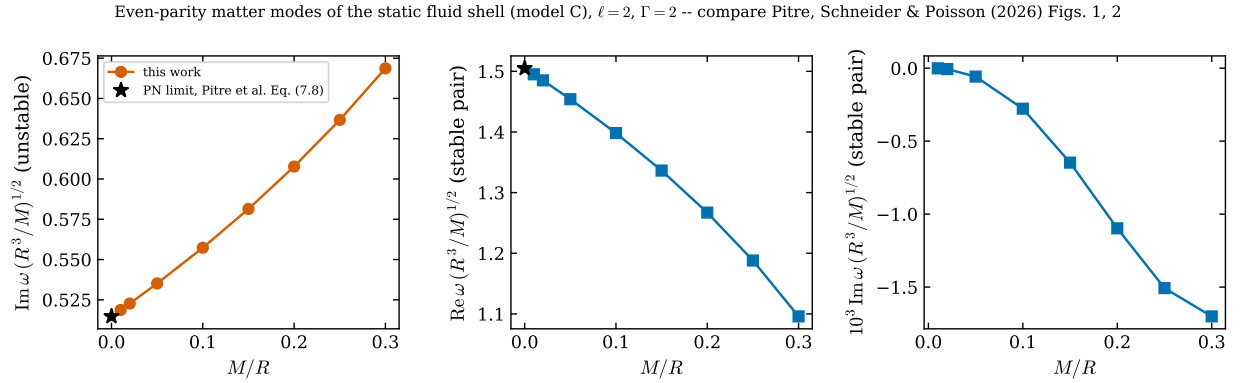


Figure 10: Matter modes of the static fluid shell vs compactness, with the post-Newtonian limits of PSP26 (stars).

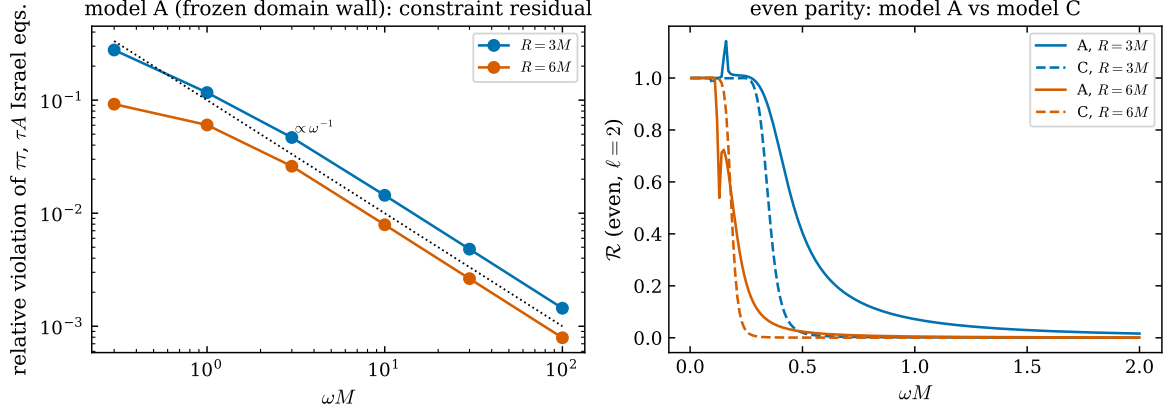


Figure 11: Model A (frozen domain wall). Left: relative violation of the $\tau\tau$ and τA Israel constraints, $\propto \omega^{-1}$. Right: even-parity reflectivity of models A and C.

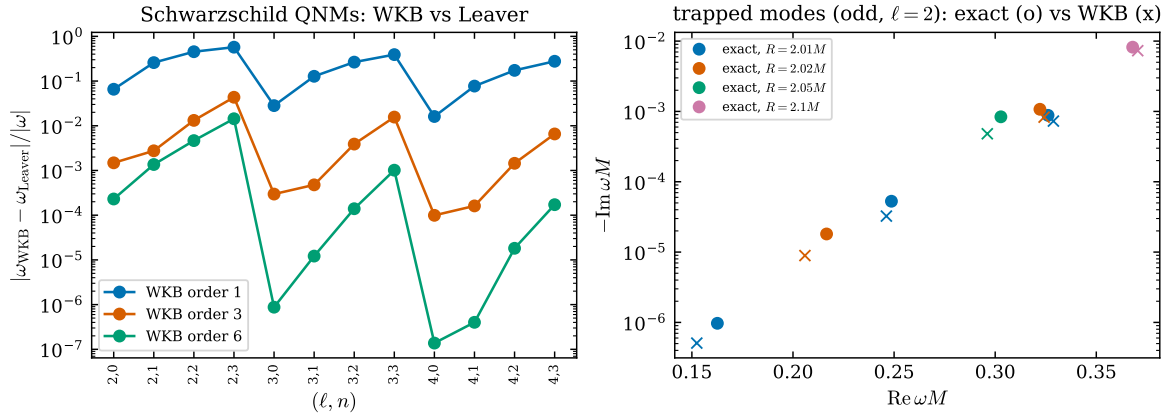


Figure 12: WKB validation. Left: Schwarzschild QNMs at WKB orders 1, 3 and 6 vs Leaver. Right: trapped modes of ultracompact shells, exact (o) vs WKB (x).